Multiple Paths to Pressure: Defensive-Front Talent Structure and Resource Allocation in the NFL

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Abstract

Consistent quarterback pressure is associated with stronger pass-defense performance, but the players most capable of generating it are scarce and expensive. This study examines defensive-front construction as a constrained roster-allocation problem: how pressure relates to defensive outcomes, how leading-player and supporting-cast quality combine to produce pressure, and whether salary-cap and draft-resource allocation reliably predicts subsequent performance. Using NFL team- and player-level data from 2022–2025, the analysis links team pressure to pass-defense outcomes, decomposes defensive-front quality into headliner and supporting creation, and evaluates lagged resource allocation, continuity, concentration, and demonstrated room quality as predictors of subsequent pressure.

Higher pressure was associated with stronger defensive EPA and success-rate performance, but the relationship was better characterized as a gradient than as a universal threshold. Four-man pressure efficiency provided the clearest personnel-linked pathway to defensive quality, while five-plus-rusher usage offered little independent explanatory value. At the room level, both leading-player and complementary creation were positively associated with pressure. A one-standard-deviation increase in headliner quality corresponded to 0.96 percentage points of overall pressure and 0.90 points of four-man pressure; corresponding support-quality estimates were 1.86 and 1.90 points.

Resource quantity and concentration were weak direct predictors of subsequent pressure. Once demonstrated prior room quality and the full room-construction controls were included, additional total resource commitment provided little incremental information, while room quality remained strongly associated with both overall and four-man pressure. That relationship also survived a player-adjusted persistence check in which previous team pressure was residualized against prior headliner and support quality; the residual prior-pressure term retained independent signal for overall pressure but not for four-man pressure. Retrospective 2025 validation similarly favored room-quality-based models for differentiating future four-man pressure performance.

These findings do not identify a single optimal defensive-front allocation structure. Instead, they reveal the central roster-building tension that emerges once quality has been demonstrated: teams must decide how much scarce cap and draft capital to commit to preserving proven creation, how much complementary quality is necessary around their best players, and which form of dependency they are willing to accept. Effective construction is therefore less a problem of selecting one ideal spending shape than of converting resources into sufficient demonstrated quality to sustain a credible four-man pressure pathway while managing the failure mode created by that choice.

1. Introduction

Consistent quarterback pressure is valuable, but the players most capable of producing it are scarce and expensive. NFL teams must therefore construct defensive fronts under competing salary-cap and draft constraints, with every investment in pass rush limiting resources available elsewhere on the roster.

Teams can approach that problem through different structures. One room may center on an elite creator supported by lower-cost contributors. Another may distribute production across several above-average rushers. Similar pressure rates can therefore emerge from rooms with different costs, dependencies, and exposure to injury, decline, or developmental failure.

This creates an important distinction between pressure production and pressure construction. Team pressure rate describes the output of the defensive front and the broader defensive system. It does not identify whether that pressure depended on one player, credible complementary production, additional rushers, or coordinated design. Nor does it establish whether the room's production is likely to persist.

The organizational problem is sequential. Teams allocate cap space and draft capital to acquire and retain players. Those players form a room with a particular distribution of demonstrated quality. That room generates pressure through four-man and additional-rusher pathways, which in turn affect the defense's range of possible outcomes. Prior work provides foundations for each stage: expected-points models connect plays to game value; tracking and charting research measures pass-rush disruption beyond sacks; and draft and salary-cap studies examine personnel allocation under uncertainty (Yurko, Ventura, and Horowitz 2019; Nguyen, Yurko, and Matthews 2024; Hong et al. 2024; Massey and Thaler 2013; Borghesi 2008; Mulholland and Jensen 2019).

This study evaluates that sequence through three analytical layers. Layer 1 estimates the relationship between pressure and pass-defense performance and distinguishes pressure produced with four rushers from pressure produced with five or more. Layer 2 examines how leading-player quality and supporting-cast quality combine within the defensive front. Layer 3 tests whether resource quantity and concentration reliably predict subsequent pressure once prior room quality, continuity, and uncertainty are considered.

The analysis addresses three questions:

1. How is pressure associated with pass-defense performance, and which rush-count pathways carry the clearest relationship?
2. How do top-end and complementary player quality combine within effective defensive-front rooms?
3. Do cap and draft resources directly predict pressure, or does demonstrated room quality provide more useful information than the nominal structure of investment?

The contribution is not an estimate of one optimal defensive-front budget. It is an integrated framework connecting resource inputs, room-level talent structure, pressure capability, and defensive outcomes.

The findings support a conditional construction model. Pressure is associated with stronger EPA and success-rate performance, and four-man pressure provides the clearest personnel-linked pathway. Both leading-player and support quality contribute to pressure production. Teams reach strong outcomes through star-led and distributed structures, but resource quantity and concentration are weak direct predictors. Once demonstrated room quality is known, additional resource commitment adds little incremental information about subsequent pressure. Demonstrated room quality also remains informative after accounting for prior team pressure in a player-adjusted persistence specification, particularly for subsequent four-man pressure.

The practical question is therefore not simply whether to pay a star or accumulate depth. It is whether the selected contracts, draft investments, and retained players create enough demonstrated quality to sustain a credible four-man pressure pathway.

2. Literature Review and Conceptual Framework

2.1 Pressure and defensive performance

Expected-points frameworks provide a common scale for evaluating football outcomes and have become central to play-by-play analysis. Yurko, Ventura, and Horowitz (2019) demonstrate both the value of EPA-based evaluation and the difficulty of isolating individual contribution in an interdependent sport.

Pressure is an intermediate event within that sequence. It occurs more frequently than sacks and may alter timing, movement, decision-making, or throw quality without producing a terminal box-score outcome. Tracking-based work has accordingly moved toward continuous or within-play measures of disruption. STRAIN measures how rapidly a defender closes distance to the quarterback, while Hong et al. (2024) develop a unified system for identifying rushers, blocker-rusher matchups, and pressure throughout a play (Nguyen, Yurko, and Matthews 2024; Hong et al. 2024).

Pressure is not a complete model of pass defense. Coverage, protection, quarterback behavior, and route structure affect both whether pressure arrives and what happens afterward. The present study therefore treats pressure as an enabling defensive capability and evaluates its relationship with EPA, success-rate prevention, and explosive-pass prevention separately.

2.2 Individual creation and room-level quality

Individual pass-rush performance is contextual. Rushers face different blockers, protection schemes, alignments, quarterbacks, and time-to-throw environments. Interaction-level work has shown the importance of opponent identity, double teams, and outcome severity when evaluating trench play (Byanna and Klabjan 2016; Pipping-Gamon et al. 2026).

The present study does not reproduce a tracking-based matchup model. It combines stabilized pressure rate, pass-rush win rate, true-pass-set pressure rate, and true-pass-set win rate to estimate demonstrated creation. Finishing is measured separately because the conversion of pressure into sacks or hits depends partly on quarterback movement, teammate interaction, and play geometry.

Most public pass-rush metrics remain player-centered. Roster decisions, however, concern an interdependent room. A dominant rusher may attract protection and create favorable opportunities for teammates. A strong supporting cast may prevent the offense from directing disproportionate attention toward one threat. Two rooms with similar aggregate pressure may therefore differ substantially in concentration, replacement difficulty, and resilience.

This study represents those dimensions through headliner and supporting-cast quality. Four transparent archetypes - Star + Depth, Heliocentric Star, Distributed Rotation, and Thin / Mixed - summarize the continuous measures. They describe observed talent structures rather than organizational intent or universally optimal models.

2.3 Resource allocation under uncertainty

NFL defensive-front construction occurs through two principal resource systems: the salary cap and the draft. Both are constrained, but they carry different forms of uncertainty.

Draft research, most notably Massey and Thaler (2013), shows that premium selections do not eliminate evaluation risk and may be priced above the surplus value they historically produce. A high draft pick represents meaningful organizational commitment, but not established NFL quality.

Salary-allocation research has examined compensation dispersion, positional investment, and constrained roster optimization. This literature treats roster construction as a portfolio problem while highlighting a central identification challenge: salary is not randomly assigned. Better players tend to receive larger contracts, while cap accounting, market scarcity, competitive timelines, and organizational evaluation also influence observed spending (Borghesi 2008; Mondello and Maxcy 2009; Jeffords and Potts 2019; Mulholland and Jensen 2019).

The present study therefore separates resource commitment, resource concentration, and realized quality. A weak direct relationship between resources and pressure would not imply that teams can build without investment. It would indicate that similar resource totals can convert into materially different levels of player quality.

Concentration and continuity are treated similarly. High concentration may represent rational investment in a rare creator or excessive dependence on one player. Low concentration may represent a deep rotation or a collection without a difference-maker. Returning production may preserve effective quality or maintain an inadequate room. Neither allocation shape nor continuity is assumed to be beneficial independently of the players involved.

2.4 Research gap and analytical framework

Existing research establishes that pressure contains information beyond sacks, that individual pass-rush performance is contextual, and that cap and draft resources involve substantial allocation uncertainty. These strands are less often connected at the defensive-front room level.

The present framework links them in the following analytical order:

Resources -> room quality and continuity -> pressure capability -> defensive outcomes

The sequence is not interpreted causally. Scheme, protection, coverage, health, development, and organizational quality may affect multiple stages. Figure 1 summarizes this analytical sequence and the contextual influences that may operate across stages.

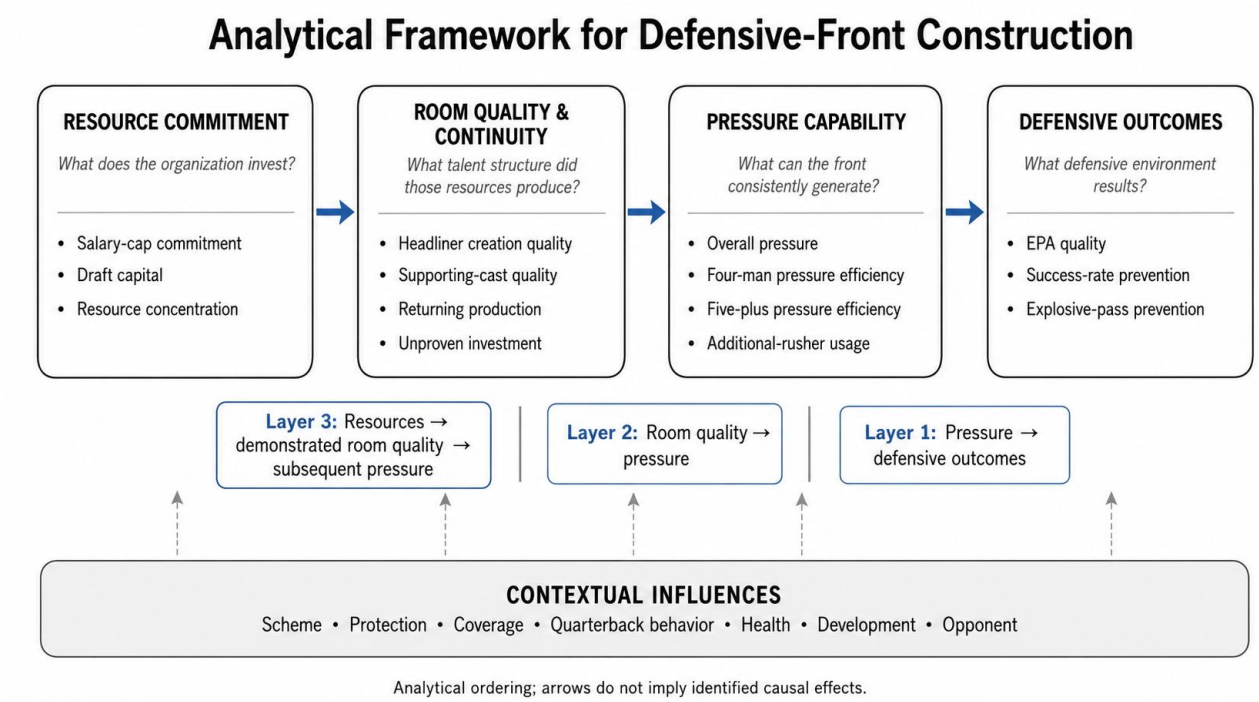


Figure 1. Analytical framework for defensive-front construction.

The empirical expectation is that demonstrated player quality and retained production will be more informative than spending amount or concentration considered in isolation. The analysis tests that expectation through continuous measures first and uses archetypes only to describe the observed construction pathways.

3. Data, Sample Construction, and Measurement

3.1 Study design and samples

This study uses a three-layer observational design. Layer 1 evaluates the relationship between team pressure and pass-defense performance. Layer 2 examines how pressure production is distributed between the room's leading creator and the supporting rotation. Layer 3 evaluates whether cap and draft resources predict subsequent pressure once demonstrated quality, continuity, and resource distribution are considered.

The primary study period is 2022-2025. Layers 1 and 2 include all 32 defenses in each season, producing 128 defense-seasons. Layer 3 requires previous-season player-quality information and begins in 2023. Its primary sample contains 82 team-seasons from 2023-2025 that satisfy the frozen resource-coverage requirement.

The design is descriptive and associational. Layer 3 preserves temporal ordering by relating prior player quality to current resources and pressure, but the lag structure does not identify a causal return to spending, draft investment, continuity, or concentration.

3.2 Data sources and defensive-front scope

Team pressure, rush-count variables, and play-level defensive outcomes were obtained from FTN play-by-play and team-defense data (FTN Data 2026). Eligible dropbacks include pass attempts, sacks, and quarterback scrambles; nullified plays were excluded. Across the four seasons, the Layer 1 sample contains 80,719 eligible dropbacks.

Player-level pass-rush opportunities, pressures, sacks, hits, hurries, pass-rush wins, true-pass-set production, position, and team assignment were obtained from Pro Football Focus (Pro Football Focus 2026). The primary defensive-front room includes players classified as edge defenders or defensive interior defenders. Pressure produced by other positions is included in team pressure rates but not in the player-based room-quality measures.

Because FTN and PFF define pressure differently, the analysis evaluates concordance rather than requiring one-for-one reconciliation. Across 128 defense-seasons, the within-season standardized PFF ED/DI room pressure rate correlated 0.724 with FTN overall pressure and 0.718 with FTN four-man pressure; corresponding Spearman rank correlations were 0.694 and 0.689 (all $p < .001$). The two provider systems therefore show strong team-season agreement while retaining distinct analytical roles.

Salary-cap commitment was reconstructed from player-season Spotrac cap-hit records frozen on July 27, 2026 (Spotrac 2026). Matched player cap hits were summed within each defensive front and divided by the official league salary cap for that season. Because cap matches covered different shares of each room's pass-rush opportunities, the primary cap measure adjusted the observed room cap share by opportunity coverage before standardization within season.

Draft commitment was measured using the Fitzgerald-Spielberger draft-value curve published by Over The Cap (Fitzgerald and Spielberger n.d.). Each pick's value was annualized over four seasons and counted while a homegrown player remained within seasons zero through three of his rookie-contract window. Players acquired from another organization were not assigned their original team's draft expenditure. The primary combined resource index equally weighted the within-season standardized cap and draft components and then standardized the resulting average within season. All source data and derived analytical files were frozen on July 27, 2026.

Traded player-seasons were represented as team-specific segments where necessary so that production, opportunity, and resources remained aligned with the team for which they occurred. Team abbreviations, player identities, opportunity totals, cap matches, and team-season uniqueness were audited before estimation.

3.3 Pressure pathways and defensive outcomes

Pressure rates were calculated overall and conditional on the number of pass rushers. Four-man pressure is the primary construction pathway because it measures disruption generated while preserving seven potential coverage

defenders. Five-plus pressure efficiency and five-plus usage are analyzed separately to distinguish the effectiveness of additional rushers from the frequency with which they are deployed.

The primary outcome is EPA allowed per eligible dropback, sign-oriented so that larger values represent better defense. Expected-points methods place plays on a common scoring-value scale (Yurko, Ventura, and Horowitz 2019). Secondary outcomes include success-rate prevention and explosive-pass prevention. An explosive pass is defined as a pass play gaining at least 20 yards.

Because league pressure rates varied by season, the main pooled Layer 1 models use within-season pressure percentile. Raw rates are retained for descriptive summaries and football interpretation.

3.4 Player creation, finishing, and room structure

The final canonical Layer 2 player-quality table contains 1,709 player-team-season records. Creation is constructed from pressure rate, pass-rush win rate, true-pass-set pressure rate, and true-pass-set win rate. Components are stabilized toward season-position priors, standardized within season and position, and combined with a volume term that separates full-time production from small-sample efficiency.

Finishing is measured separately through stabilized conversion of pressures into sacks or quarterback hits. It does not alter the creation ranking because sacks and hits occur downstream from pressure and are more sensitive to quarterback behavior, teammate interaction, and play geometry.

The room's headliner is the player with the highest creation score. Support quality is the opportunity-weighted creation quality of the remaining qualifying players. Four rule-based archetypes summarize these continuous measures: Star + Depth, Heliocentric Star, Distributed Rotation, and Thin / Mixed. The categories are descriptive rather than estimated latent classes.

3.5 Resource allocation and prior room quality

Total resource commitment is represented by the frozen combined index of salary-cap and annualized draft inputs. A separate positive player-resource scale is used to study how those resources are distributed within the room.

Each player's within-room resource share equals the player's resource units divided by total room resource units. Resource concentration is measured primarily through the Herfindahl-Hirschman Index, the sum of squared player resource shares (Rhoades 1993). Additional descriptive measures include leading-player share, the combined share assigned to players two and three, and effective resource-player counts.

The primary prior-room-quality measure is the equal-weight mean of the previous-season creation percentiles of current resource-bearing players. Players without a valid previous-season estimate receive a neutral 50th-percentile value; their resource share is tracked separately as unproven investment. For interpretation, room quality is scaled so that one regression unit equals 10 percentile points.

Resource-weighted prior quality weights each player's prior percentile by current resource share. Alignment is the difference between resource-weighted and equal-weight prior quality. Continuity is measured through returning prior rush share, and previous team pressure is created by shifting each defense's overall and four-man pressure rates forward one season. Because raw previous pressure partly reflects persistent player ability, an additional sensitivity measure residualizes each prior season's team pressure against that season's headliner and support quality using leave-one-team-out prediction within season; the resulting observed-minus-expected residual is then shifted forward as a player-adjusted persistence control.

3.6 Data integrity and reproducibility

The final Layer 3 specification was rebuilt from retained canonical inputs in a fresh computational environment. The pipeline reconstructs player prior quality, within-room resource shares, room quality, alignment, unproven share, concentration, and lagged pressure without relying on inherited notebook objects.

The peer-review sensitivity analyses were implemented as standalone scripts against the same frozen inputs, producing separate derived outputs for the conditional resource test, the player-adjusted prior-pressure residual, and the FTN-PFF concordance audit.

The resulting locked panel contains 82 unique team-seasons. A machine-readable manifest records source paths, checksums, transformation rules, missing-prior treatment, and sample dimensions. The final panel was reloaded independently and audited for completeness, uniqueness, bounded shares, and the expected 55-observation training and 27-observation holdout split.

4. Empirical Strategy and Statistical Methods

4.1 Layer 1 models

Layer 1 combines descriptive comparisons with pooled regression models. Ordinary least squares relates sign-oriented EPA quality to within-season pressure percentile and season indicators. Logistic generalized linear models estimate the probability of reaching average-or-better and good-or-elite defensive outcomes. Linear and spline specifications are compared to evaluate whether the relationship supports a distinct pressure threshold.

A 5,000-draw bootstrap estimates coefficient uncertainty and the change in defensive quality associated with movement across the interquartile range of pressure (Efron and Tibshirani 1993). The pressure-pathway model relates defensive outcomes simultaneously to four-man pressure efficiency, five-plus pressure efficiency, and five-plus usage. Removal-based changes in R-squared compare their explanatory contributions.

4.2 Layer 2 models

Layer 2 uses season-adjusted ordinary least squares to estimate the simultaneous relationships between headliner quality, support quality, and team pressure. Overall and four-man pressure are modeled separately. Archetype comparisons summarize the same underlying quality dimensions and are interpreted descriptively.

Creation and finishing are evaluated separately, and the continuous measures carry the primary inferential weight. Threshold-based archetype contrasts are used to describe observed construction pathways rather than to estimate a universal taxonomy.

4.3 Layer 3 explanatory models

Layer 3 uses nested team-season regressions with season indicators and team-clustered standard errors. Cluster-robust inference accounts for repeated observations of the same organizations (Cameron and Miller 2015). The models are entered sequentially: resources and concentration; prior room quality; continuity and unproven share; and alignment.

Overall and four-man pressure are expressed in percentage points. Resource commitment and HHI are standardized for coefficient comparability. Room quality, alignment, unproven share, and continuity are entered on scales where one unit corresponds to 10 percentile points or 10 percentage points, as applicable.

Separate models evaluate total resources, HHI, leading-player resource share, and the combined share assigned to players two and three. Interaction specifications test whether concentration moderates the relationship between prior room quality and subsequent pressure. A targeted conditional-resource sensitivity adds a total-resource-index by prior-room-quality interaction to the full room model and evaluates whether the incremental resource association changes across the observed room-quality distribution.

Temporal-persistence models compare previous team pressure with prior player-based room quality. The baseline sequence includes prior pressure alone, room quality alone, both jointly, the complete room-construction model, and the complete model with prior pressure added. To address overlap between raw prior pressure and persistent player ability, a sensitivity specification first predicts each prior-season overall and four-man pressure rate from that season's headliner and support quality using leave-one-team-out estimation within season.

Observed-minus-expected residual pressure is then shifted forward and substituted for raw prior pressure in the persistence models.

4.4 Retrospective temporal validation

Models were estimated using the 55 team-seasons from 2023-2024 and evaluated on the 27 qualifying defenses from 2025. All preprocessing parameters were estimated on the training sample. Season indicators were excluded because a 2025 effect cannot be estimated from the training seasons.

Five prespecified models were evaluated for overall and four-man pressure: prior pressure; prior room quality; the two jointly; the complete room model; and the complete room model plus prior pressure. Performance is evaluated using MAE, RMSE, Pearson correlation, mean signed error, conventional test-set R-squared, and out-of-sample R-squared relative to the training-sample mean.

The exercise is described as retrospective or pseudo-out-of-sample validation. The 2025 observations were excluded from estimation and preprocessing during the validation step, but the broader variable framework was developed with knowledge of the full study period.

4.5 Robustness and claim discipline

Robustness analyses include alternative outcomes, uncertainty estimators, draft-value constructions, cap-coverage thresholds, leave-one-season-out and leave-one-team-out estimation, influence diagnostics, alternative concentration measures, and alternative room-quality constructions. Additional peer-review checks evaluate conditional resource-by-quality effects, player-adjusted residual prior-pressure controls, and FTN-PFF cross-provider concordance.

Statistical significance is not treated as a substitute for practical magnitude. Conclusions are based on coefficient size, uncertainty, stability across specifications, consistency between overall and four-man outcomes, and temporal validation. Null estimates are interpreted as a lack of reliable evidence in the observed sample rather than proof that a mechanism can never matter.

5. Results I: The Pressure Requirement

5.1 League-wide pressure environment

The league pressure environment changed during the study period. Mean overall pressure rate increased from 27.14% in 2022 to 29.50% in 2023 and 30.88% in 2024 before settling at 29.76% in 2025. Median four-man pressure rose from 25.10% in 2022 to 29.33% in 2025, while median five-plus usage declined from 25.95% to 21.56%.

The trend does not establish a causal league-wide movement toward four-man pressure, but it illustrates why a fixed raw benchmark is difficult to interpret across seasons. The primary pooled models therefore use within-season pressure percentiles or season indicators.

5.2 Pressure and pass-defense quality

Pressure increased monotonically across the five defensive EPA tiers. Median pressure rates were 26.97% for Poor defenses, 28.66% for Below Average defenses, 29.43% for Average defenses, 30.49% for Good defenses, and 32.08% for Elite defenses.

In the pooled OLS model, within-season pressure percentile was positively associated with EPA quality. The coefficient was 0.117 ($p < .001$), and the model explained 26.1% of the variation in the defensive outcome. A 5,000-draw bootstrap estimated that movement across the interquartile range of pressure was associated with a 0.0587 increase in EPA quality, with a 95% confidence interval from 0.039 to 0.077.

Pressure explains a portion of pass-defense quality, not the entire defense. Coverage, quarterback quality, game state, tackling, turnovers, and explosive-play prevention remain jointly involved in the observed outcome. Full Layer 1 model estimates are reported in Appendix Table A1.

5.3 Gradient rather than threshold

The probability models support a gradient more strongly than a universal threshold. The estimated pressure percentile at which a defense crossed a 50% probability of being average or better was approximately the 33rd percentile, with a wide 95% confidence interval from the 13th to the 47th percentile. The corresponding crossing point for a good-or-elite defense was approximately the 66th percentile, with a confidence interval from the 50th to the 86th percentile.

Spline specifications did not improve fit significantly relative to the linear relationship. The crossing points are therefore best treated as viability references rather than necessary or sufficient benchmarks. Figure 2 displays the observed pressure distributions across EPA-quality tiers together with the modeled probability gradients.

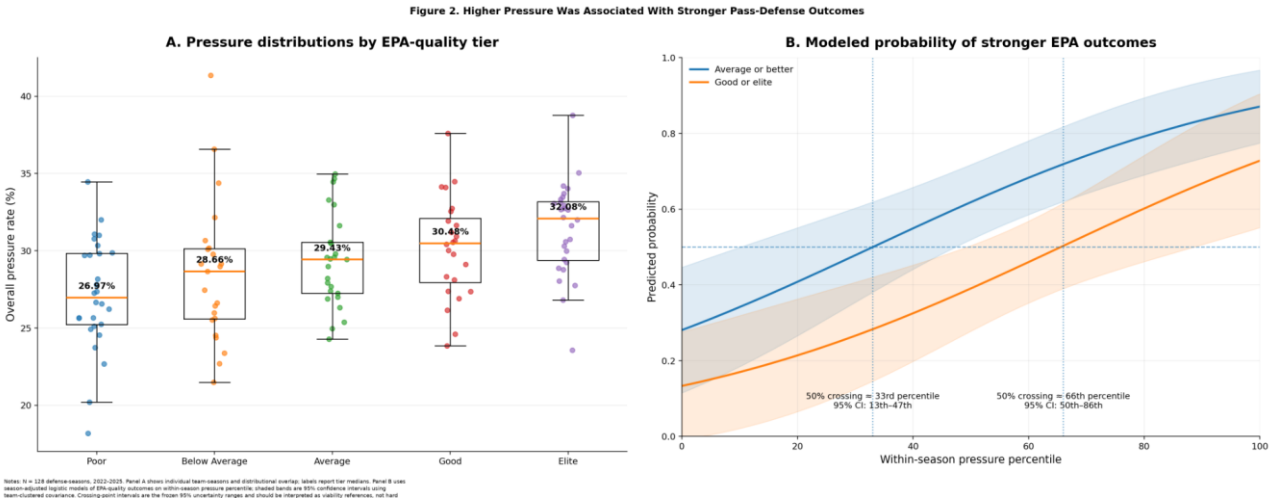


Figure 2. Pressure distributions by EPA-quality tier and modeled probabilities of stronger defensive outcomes.

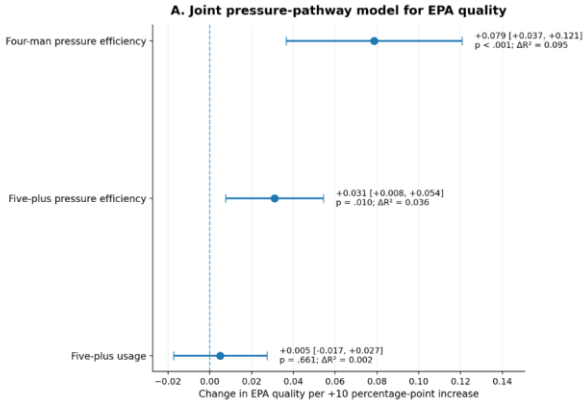
5.4 Four-man and additional-rusher pathways

In the joint pathway model, four-man pressure efficiency carried a coefficient of 0.787 ($p < .001$), while five-plus pressure efficiency carried a smaller positive coefficient of 0.310 ($p = .010$). Five-plus usage itself was not statistically distinguishable from zero.

The full model explained 29.3% of the variation in EPA quality. Removing four-man efficiency reduced R-squared by 0.095, compared with 0.036 when five-plus efficiency was removed. Secondary models showed a similar directional pattern for success-rate prevention. Neither pathway reliably predicted prevention of passes gaining at least 20 yards.

The evidence does not imply that additional rushers are inherently ineffective. Five-plus pressure can be productive when it generates pressure efficiently. The distinction is between pressure efficiency and pressure frequency. Figure 3 summarizes the joint pathway estimates and the corresponding outcome-boundary checks.

Figure 3. Four-Man Pressure Efficiency Was the Clearest Personnel-Linked Pathway



B. Outcome boundaries across pressure pathways

Pressure pathway	EPA quality	Success-rate prevention	20+ yd explosive-pass prevention
Four-man pressure efficiency	**+0.079** p < .001	**+1.86 pp** p = .021	+0.21 pp p = .528
Five-plus pressure efficiency	**+0.031** p = .010	+1.28 pp p = .057	-0.08 pp p = .687
Five-plus usage	+0.005 p = .661	+0.17 pp p = .693	+0.06 pp p = .799
Full-model R^2:	**0.293**	**0.150**	**0.012**

Effects are shown for a +10 percentage-point increase in each pressure variable. For success-rate and explosive-pass prevention, effects are percentage points. Bold cells have p < .05 using team-clustered standard errors.

Notes: N = 128 defense-seasons, 2022–2025. All models include season fixed effects and team-clustered standard errors. Panel A reports the joint EPA-quality model with 95% confidence intervals and the reduction in model R^2 when each pathway is removed. Panel B shows the same three predictors across EPA quality, success-rate prevention, and prevention of pass plays gaining at least 20 yards. Higher outcome values indicate better defense. Estimates are associational.

Figure 3. Pressure pathways and outcome boundaries across four-man efficiency, five-plus efficiency, and five-plus usage.

5.5 Section synthesis

Strong pass defense does not require one exact pressure threshold, but access to better outcomes becomes increasingly likely as pressure rises. Four-man pressure is the clearest personnel-linked route because it creates disruption without requiring an additional rusher.

The roster implication is not simply to maximize raw pressure. It is to build a front with a credible four-man floor while retaining the ability to supplement that floor selectively.

6. Results II: Defensive-Front Talent Structures

6.1 Headliner and supporting-cast quality

Headliner and support quality were each positively associated with team pressure when entered simultaneously. A one-standard-deviation increase in headliner quality was associated with 0.96 percentage points of overall pressure and 0.90 points of four-man pressure. The corresponding estimates for support quality were 1.86 and 1.90 points.

These estimates address different dimensions of the room. Headliner quality measures its best creator; support quality aggregates the remaining rotation. The results indicate that leading-player quality raises expected pressure, but does not substitute consistently for quality elsewhere in the room.

The support relationship was similarly strong for four-man pressure. Complementary creation was therefore associated not only with more total pressure, but with pressure generated without routinely adding rushers. The continuous headliner and support estimates are summarized in Figure 4A; full model results are reported in Appendix Table A2.

6.2 Room archetypes

The continuous quality measures were summarized through four predefined archetypes. Star + Depth rooms combined a qualifying creation star with support quality at or above the frozen 60th-percentile boundary. Distributed Rotation rooms contained no qualifying star but included strong collective support and at least three credible rushers. Heliocentric Star rooms paired a qualifying star with support below the frozen boundary. Thin / Mixed rooms met none of the other definitions.

The structural profiles confirm that the categories captured distinct distributions of talent and production. The categories describe talent structures, not spending strategies or organizational intent. Frozen archetype definitions and structural profiles are reported in Appendix Table A3.

6.3 Pressure production across archetypes

Median overall pressure followed the expected ordering. Star + Depth recorded the highest median pressure rate at 32.62%, followed by Distributed Rotation at 30.69%, Heliocentric Star at 29.35%, and Thin / Mixed at 26.89%.

Star-led rooms outperformed the thin baseline, indicating that top-end creation matters. Star + Depth outperformed Heliocentric Star, indicating that support remained relevant within star-led rooms. Distributed Rotation produced the strongest non-star result, demonstrating that several credible creators can provide an alternative pathway.

The categories simplify variation within each group. Their value is narrower: strong pressure was reached through more than one talent structure, while a star without sufficient support produced less complete results than a star embedded in a credible rotation. The observed archetype distributions are shown in Figure 4B.

Figure 4. Headliner quality lifts pressure, but support and room structure separate the best rooms.

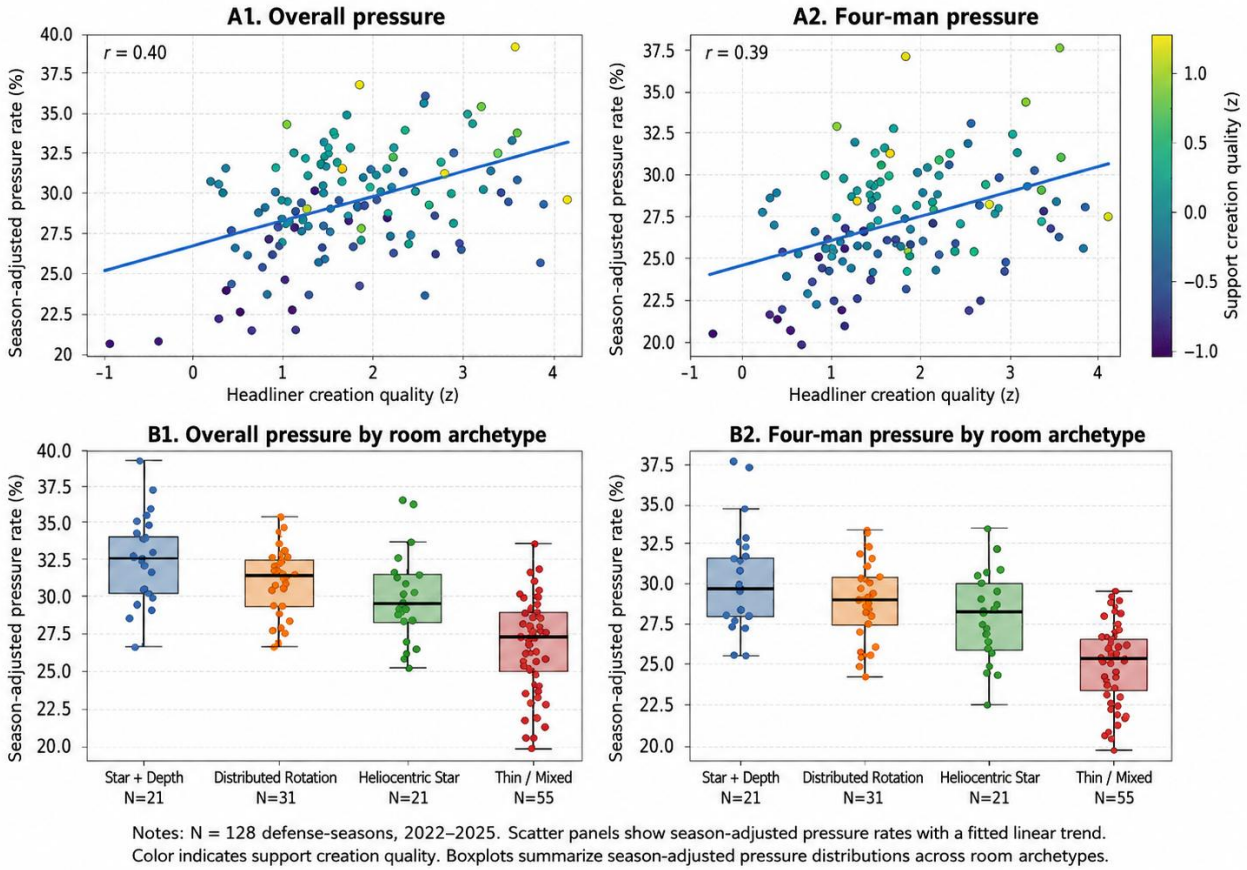


Figure 4. Headliner quality, support quality, and pressure distributions across defensive-front room archetypes.

6.4 Creation and finishing

Creation and finishing were retained as separate traits. Creation measures how frequently a player produces disruption across pass-rush opportunities. Finishing measures how frequently those pressures become sacks or quarterback hits.

Finishing helped distinguish players who converted disruption into terminal outcomes, but it did not replace creation as the primary room-level measure. Conversion is affected by pressure timing, quarterback behavior, escape paths, and teammate interaction. The room-construction analysis therefore centers on the process that must

be reproduced - pressure creation - rather than sack totals alone. Year-over-year stability of creation and finishing measures is documented in Appendix Figure A1.

6.5 Persistence of room quality

Lagged models tested whether room quality contained information beyond the defense's own previous pressure rate. For overall pressure, adding prior room quality to a model containing previous team pressure increased adjusted R-squared from 0.173 to 0.243. In the joint model, a 10-percentile-point increase in prior room quality was associated with 1.36 percentage points of subsequent pressure ($p = .015$). Previous overall pressure also retained independent signal.

The distinction was stronger for four-man pressure. Adding room quality increased adjusted R-squared from 0.102 to 0.182. A 10-percentile-point increase in room quality was associated with 1.42 percentage points of subsequent four-man pressure in the joint model ($p = .016$).

In the fuller room-construction specification, the four-man room-quality estimate remained positive at 1.69 points ($p = .010$), while previous team four-man pressure was no longer independently significant ($p = .220$). These raw-pressure specifications provide the baseline persistence comparison shown in Figure 5.

Because raw prior pressure can contain the persistent ability of returning players, a leave-one-team-out sensitivity removed the portion of previous-season pressure predicted by same-season headliner and support quality. With that player-adjusted control, a 10-percentile-point increase in prior room quality remained associated with 2.02 percentage points of subsequent overall pressure (95% CI [0.99, 3.05], $p < .001$) and 1.84 points of subsequent four-man pressure (95% CI [0.80, 2.88], $p < .001$). Residual prior overall pressure retained independent signal (0.294, $p = .011$), while residual prior four-man pressure did not (0.106, $p = .215$).

In the full room-construction models, the result was similar: prior room quality remained associated with 2.18 points of overall pressure ($p < .001$) and 2.06 points of four-man pressure ($p < .001$). The residual prior-pressure term remained significant for overall pressure (0.244, $p = .036$) but not for four-man pressure (0.092, $p = .349$). The additional persistence results are reported in Appendix Table A4.

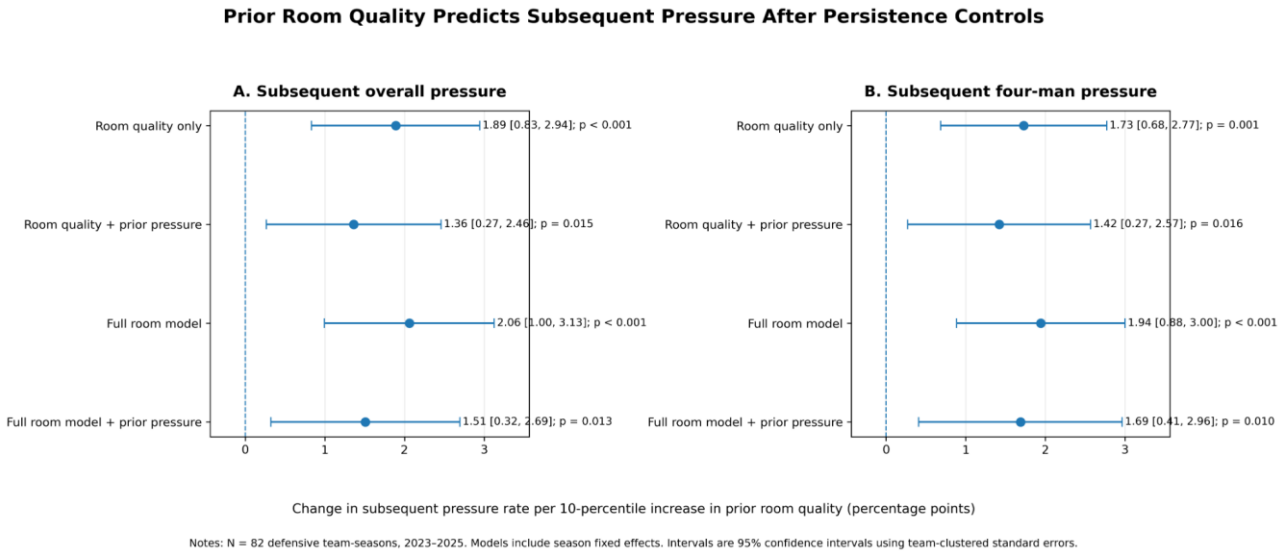


Figure 5. Prior room quality as a predictor of subsequent pressure after persistence controls.

6.6 Retrospective temporal validation

Models estimated on the 55 qualifying team-seasons from 2023-2024 were evaluated on the 27-team 2025 holdout. No 2025 observations were used to estimate coefficients or preprocessing parameters.

For four-man pressure, room quality alone reduced MAE by 8.6% and RMSE by 9.9% relative to the previous-pressure baseline. The complete room model reduced MAE by 11.2% and RMSE by 14.1%, producing an out-of-sample R-squared of 0.297 relative to the training-sample mean. Adding prior team four-man pressure did not improve that model.

Overall-pressure gains were smaller. Combining prior team pressure with room quality reduced MAE by 4.3% and RMSE by 5.2%. The complete room model produced the lowest RMSE, although it did not improve MAE relative to the persistence baseline.

The four-man models overpredicted the 2025 league environment. The holdout therefore provides stronger evidence for relative differentiation among rooms than for exact season-level pressure forecasts. Full holdout performance metrics are reported in Table 1.

Table 1. Retrospective 2025 Temporal Validation of Subsequent Pressure Models

Panel A. Overall pressure

Model	Predictors	MAE	RMSE	OOS R^2	Bias	Δ MAE vs. P0	Δ RMSE vs. P0
P0	Prior overall pressure	2.201	2.948	0.214	1.152	—	—
Q0	Prior room quality	2.266	2.861	0.260	**0.152**	2.9%	-3.0%
PQ	Prior pressure + room quality	**2.107**	2.795	0.294	0.869	**-4.3%**	-5.2%
R0	**Full room model**	**2.223**	**2.721**	**0.330**	**0.260**	**1.0%**	**-7.7%**
R1	Full room model + prior pressure	2.153	2.812	0.285	0.816	-2.2%	-4.6%

Panel B. Four-man pressure

Model	Predictors	MAE	RMSE	OOS R^2	Bias	Δ MAE vs. P0	Δ RMSE vs. P0
P0	Prior four-man pressure	2.505	3.296	0.048	1.625	—	—
Q0	Prior room quality	2.289	2.969	0.227	**1.104**	-8.6%	-9.9%
PQ	Prior pressure + room quality	2.444	3.083	0.167	1.412	-2.4%	-6.5%
R0	**Full room model**	**2.224**	**2.831**	**0.297**	**1.247**	**-11.2%**	**-14.1%**
R1	Full room model + prior pressure	2.343	2.973	0.225	1.453	-6.5%	-9.8%

Notes: Models were estimated on 55 qualifying defensive team-seasons from 2023–2024 and evaluated on 27 qualifying 2025 defenses. P0 uses previous team pressure; Q0 uses prior demonstrated room quality; PQ includes both; R0 is the complete room-construction holdout model (total resource index, prior room quality, returning rush share, unproven resource share, and resource-quality alignment); R1 adds previous team pressure to R0. Preprocessing parameters were estimated on the training sample. OOS R^2 is calculated relative to the training-sample mean. Bias = predicted minus actual, so positive values indicate overprediction. Negative MAE/RMSE changes indicate improvement relative to P0. This is retrospective temporal validation rather than a prospective test.

Table 1. Retrospective 2025 temporal validation of subsequent pressure models.

6.7 Section synthesis

Pressure creation was a room-level capability. Leading-player quality mattered, but support quality provided independent signal of similar or greater magnitude. Star + Depth produced the strongest observed profile, while Distributed Rotation established a viable non-star pathway.

Prior room quality predicted subsequent pressure beyond previous team performance and generalized to the 2025 holdout, with the clearest gains for four-man pressure. That conclusion also survived a player-adjusted persistence control designed to remove the component of previous pressure attributable to demonstrated headliner and support quality. The next question is whether cap and draft allocation reliably produce that quality.

7. Results III: Resource Allocation and Defensive-Front Construction

Layer 3 separates three concepts: how much a team invested, how narrowly those resources were distributed, and the demonstrated quality of the players receiving them. The primary sample contains 82 team-seasons from 2023-2025 meeting the predefined 80% opportunity-weighted cap-coverage threshold.

7.1 Allocation landscape

Defensive-front construction varied substantially in both total resource commitment and concentration. Teams reached similar levels of investment through different combinations of cap and draft resources, and those resources were distributed across rooms ranging from highly concentrated to broadly allocated.

High- and low-quality rooms appeared across each part of the allocation landscape. Neither high investment nor a particular level of concentration isolated the strongest rooms. Resource amount and resource distribution did not identify the quality attached to those resources. The allocation landscape is shown in Figure 6A.

7.2 Direct resource associations

Separate season-adjusted models evaluated total resources, resource HHI, leading-player share, and the combined share assigned to players ranked second and third.

None produced a stable direct relationship with overall or four-man pressure. All eight 95% confidence intervals crossed zero. The estimates did not support a general rule that more resources, greater concentration, or broader distribution independently generated more pressure.

The result does not imply that resources are irrelevant. Cap space and draft capital are required to acquire and retain pass-rush talent. The finding is that their nominal quantity and distribution did not determine the production subsequently observed. Common-model estimates for the primary room-construction predictors are shown in Figure 6B.

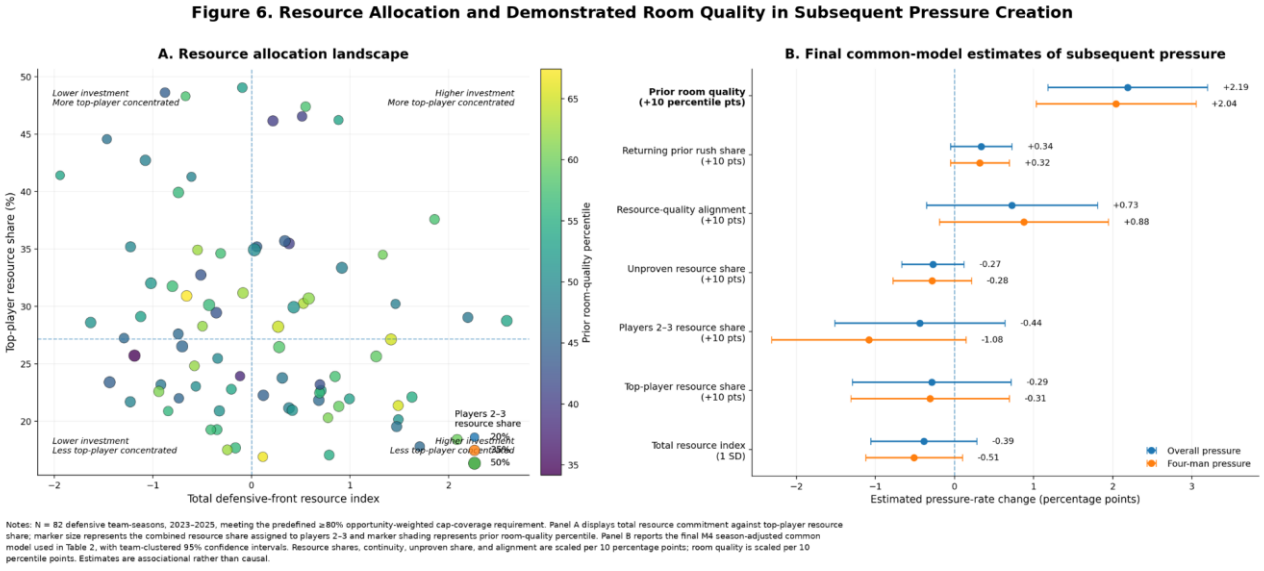


Figure 6. Resource allocation and demonstrated room quality in subsequent pressure creation.

7.3 Nested resource and room-quality models

A model containing resource quantity and concentration produced adjusted R-squared values of -0.022 for overall pressure and 0.008 for four-man pressure. The resource-allocation coefficients were not distinguishable from zero.

Adding prior room quality increased adjusted R-squared to 0.150 for overall pressure and 0.153 for four-man pressure. A 10-percentile-point increase in prior room quality was associated with 1.99 percentage points of overall pressure and 1.83 points of four-man pressure.

Those estimates remained stable as continuity and unproven resource share were added. In the final specifications, room-quality estimates were 2.19 points for overall pressure and 2.04 points for four-man pressure. The resource and concentration coefficients remained weak. The nested specifications are reported in Table 2.

A targeted conditional-resource analysis sharpened that result. In the full room models, the total resource index was -0.39 percentage points for overall pressure (95% CI [-1.06, 0.28], p = .255) and -0.51 points for four-man pressure (95% CI [-1.12, 0.10], p = .100), while prior room quality remained strongly positive at 2.19 and 2.04 points. Resource-by-quality interactions were also imprecise for overall pressure (beta = -0.45, p = .276) and four-man pressure (beta = -0.38, p = .300). The data therefore do not identify a quality threshold at which resources cease to matter; they show that additional total resource commitment did not provide a reliable positive incremental association once demonstrated room quality was known.

Returning rush share provided secondary positive signal. Unproven resource share was directionally negative but imprecise, and resource-quality alignment was positive but imprecise.

Table 2. Nested Models of Resource Allocation, Room Quality, and Subsequent Pressure

Panel A. Subsequent overall pressure (percentage points)

Predictor	M1 Resources & concentration	M2 + Room quality	M3 + Continuity / uncertainty	M4 + Alignment
Total resource index (1 SD)	+0.17 (0.43)	-0.05 (0.39)	-0.33 (0.37)	-0.39 (0.34)
Top-player resource share (+10 pts)	-0.26 (0.56)	-0.02 (0.46)	-0.03 (0.48)	-0.29 (0.51)
Players 2–3 resource share (+10 pts)	-0.37 (0.72)	-0.17 (0.65)	-0.28 (0.60)	-0.44 (0.55)
Prior room quality (+10 percentile pts)	**—**	**+1.99*** (0.56)**	**+2.09*** (0.54)**	**+2.19*** (0.51)**
Returning prior rush share (+10 pts)	—	—	+0.36 (0.19)	+0.34 (0.20)
Unproven resource share (+10 pts)	—	—	-0.36 (0.20)	-0.27 (0.20)
Resource-quality alignment (+10 pts)	—	—	—	+0.73 (0.55)
Adjusted R^2	**-0.022**	**0.150**	**0.211**	**0.226**
N	82	82	82	82
Season fixed effects	Yes	Yes	Yes	Yes
Team-clustered SEs	Yes	Yes	Yes	Yes

Panel B. Subsequent four-man pressure (percentage points)

Predictor	M1 Resources & concentration	M2 + Room quality	M3 + Continuity / uncertainty	M4 + Alignment
Total resource index (1 SD)	+0.03 (0.40)	-0.17 (0.38)	-0.44 (0.36)	-0.51 (0.31)
Top-player resource share (+10 pts)	-0.20 (0.57)	+0.02 (0.49)	+0.00 (0.51)	-0.31 (0.51)
Players 2–3 resource share (+10 pts)	-0.96 (0.77)	-0.78 (0.70)	-0.89 (0.68)	-1.08 (0.63)
Prior room quality (+10 percentile pts)	**—**	**+1.83** (0.57)**	**+1.92*** (0.54)**	**+2.04*** (0.52)**
Returning prior rush share (+10 pts)	—	—	+0.34* (0.18)	+0.32 (0.19)
Unproven resource share (+10 pts)	—	—	-0.38 (0.24)	-0.28 (0.25)
Resource-quality alignment (+10 pts)	—	—	—	+0.88 (0.54)
Adjusted R^2	**0.008**	**0.153**	**0.219**	**0.246**
N	82	82	82	82
Season fixed effects	Yes	Yes	Yes	Yes
Team-clustered SEs	Yes	Yes	Yes	Yes

*Notes: Estimates are coefficients with team-clustered standard errors in parentheses. Models include season fixed effects and use 82 defensive team-seasons from 2023–2025 meeting the predefined ≥80% opportunity-weighted cap-coverage requirement. Total resource index is standardized; resource shares, returning prior rush share, unproven resource share, and resource-quality alignment are reported per 10-percentage-point increase; prior room quality is reported per 10-percentile-point increase. M1 includes resource quantity and concentration; M2 adds demonstrated prior room quality; M3 adds continuity and unproven-resource share; M4 adds resource-quality alignment. Alternative categorical representations of resource concentration produced similar substantive conclusions. Estimates are associational rather than causal. * $p < .05$, ** $p < .01$, *** $p < .001$.*

Table 2. Nested models of resource allocation, room quality, and subsequent pressure.

7.4 Does concentration modify the value of quality?

The direct concentration estimates do not determine whether allocation shape changes how demonstrated player quality translates into pressure. Appendix Figure A3 tests that question using rank-specific interactions between top-player resource share and prior best-player quality, and between the combined resource share of players two and three and prior support quality.

Neither interaction was reliably distinguishable from zero for overall or four-man pressure. An alternative HHI-based interaction produced the same substantive conclusion (Appendix Table A5).

The analysis therefore provides no evidence that resource concentration consistently amplified or diminished the value of demonstrated quality. Strong rooms existed under both concentrated and distributed resource structures.

7.5 Section synthesis

Resource quantity and concentration were weak direct predictors of pressure. Model fit improved only after demonstrated player quality entered the analysis, and additional total resource commitment remained uninformative once the quality already produced by the room was accounted for.

The distinction is between resource commitment and resource conversion. Teams must spend cap space and draft capital to acquire, develop, and retain talent, but similar resource structures can produce materially different rooms. Once quality has been demonstrated, the incremental roster question shifts from how much more to invest toward whether scarce resources are best used to preserve that quality, replace uncertain components, or be reallocated elsewhere. The evidence does not identify one preferred allocation shape; concentrated and distributed approaches should be evaluated by the player-quality structure they produce.

8. Robustness, Validation, and Alternative Explanations

This section evaluates whether the primary conclusions survive alternative specifications, samples, outcomes, and temporal tests. Continuous measures carry the principal inferential weight; archetypes and allocation categories remain descriptive summaries.

8.1 Stability of the primary findings

The principal Layer 1, Layer 2, and Layer 3 estimates were evaluated under alternative uncertainty estimators, sample exclusions, coverage rules, and variable constructions.

The pressure-EPA relationship remained positive in every leave-one-season-out specification. The headliner and support results remained directionally consistent when represented continuously rather than through the archetypes. Prior room quality was similarly stable across clustered and heteroskedasticity-robust uncertainty estimates, alternative coverage samples, and influence checks. The full room-quality sensitivity distribution is shown in Appendix Figure A2, with the robustness specification registry reported in Appendix Table A5.

By contrast, total resource commitment, concentration, alignment, and unproven resource share did not become stable positive predictors. In the targeted conditional-resource test, total resource commitment remained imprecise after demonstrated room quality and the full room-construction controls were included, and the resource-by-quality interaction was not distinguishable from zero for either outcome. These checks reinforce the paper's evidentiary distinction: demonstrated player quality is a primary result, while direct incremental resource-allocation effects are not. Additional peer-review sensitivity results are reported in Appendix Table A4.

Outcome checks also define the boundary of the pressure result. Pressure was associated with EPA quality and success-rate prevention, but not with prevention of passes gaining at least 20 yards. Four-man and five-plus pressure efficiency were positively associated with defensive quality; five-plus usage was not independently informative.

8.2 Temporal validation

Lagged models tested whether prior room quality contained information beyond the defense's previous pressure rate. For both outcomes, adding room quality improved explanatory fit. A player-adjusted sensitivity then residualized previous-season pressure against same-season headliner and support quality using leave-one-team-out estimation. Prior room quality remained strongly positive under that control. For four-man pressure, the residual prior-pressure term was not independently significant in either the simple joint model ($p = .215$) or the full room model ($p = .349$), while room quality remained significant in both. For overall pressure, residual prior pressure retained a smaller independent signal, consistent with broader team/system persistence beyond the ED/DI room.

The retrospective temporal holdout produced a similar pattern. Prior room quality and the complete room model both outperformed the prior-pressure baseline for four-man pressure. Overall-pressure gains were smaller and were strongest when previous pressure and room quality were combined.

The four-man models overpredicted the 2025 league environment, indicating imperfect calibration. The temporal evidence is therefore stronger for differentiating room quality than for producing exact season-level forecasts.

8.3 Measurement and reproducibility

FTN and PFF serve different analytical roles. FTN defines team pressured dropbacks, rush-count pathways, and defensive outcomes. PFF supplies player opportunities and creation measures. The analysis does not require the two providers' pressure totals to match mechanically because a single pressured dropback may generate multiple player pressure credits, FTN team pressure can include non-ED/DI rushers, and provider definitions may differ.

To quantify the measurement relationship directly, a defense-season concordance audit compared the PFF ED/DI room pressure rate with FTN team pressure. After within-season standardization, PFF room pressure correlated 0.724 with FTN overall pressure and 0.718 with FTN four-man pressure; corresponding Spearman correlations were 0.694 and 0.689 (all $p < .001$; $N = 128$). Year-by-year Pearson correlations remained positive in every season. The result supports using each provider in its intended analytical role while treating concordance, rather than numerical equality, as the appropriate reconciliation standard.

The final Layer 3 specification was rebuilt from retained canonical inputs in a fresh computational environment. The process reconstructed the player-quality-derived measures, produced the locked 82-row panel, and generated the 55-observation training and 27-observation holdout samples without relying on inherited notebook objects. The reproducibility and analytic-panel audit is summarized in Appendix Table A6.

Source checksums, transformation rules, missing-prior treatment, and sample dimensions were recorded in a machine-readable manifest. The final panel was reloaded independently and audited for completeness and uniqueness.

An earlier exploratory intermediate object could not be reconstructed exactly because it was not retained. Estimates dependent on that object are excluded from the final evidence.

8.4 Claim survival

The robustness review narrows the paper's conclusions. Findings closest to the observed sequence from player quality to pressure remain the strongest. Claims involving universal spending levels, preferred concentration structures, or broad downstream defensive effects do not survive with equal support.

The pressure gradient with EPA and success-rate quality is retained. Four-man pressure is retained as the clearest personnel-linked pathway, but not as a universally superior tactical choice. Headliner and support quality are retained as independent room-level correlates.

Star + Depth and Distributed Rotation remain viable observed pathways, while their exact thresholds and category contrasts remain descriptive. Prior room quality is retained as a durable and forward-looking measure, particularly for four-man pressure; that conclusion survives the player-adjusted residual persistence test.

Direct claims about a universal positive return to greater resource quantity or a preferred concentration structure are not retained. Conditional on demonstrated room quality, additional total resource commitment adds little incremental information about subsequent pressure in the observed sample. Continuity remains a secondary correlate. Alignment remains exploratory, and unproven resource share is interpreted as uncertainty rather than evidence of low quality.

9. Discussion

9.1 Principal findings

This study evaluated defensive-front construction as a sequence linking resource allocation, player quality, pressure generation, and defensive performance.

Pressure was associated with stronger pass-defense outcomes, but the pathway mattered. Four-man pressure provided the clearest connection between defensive-front personnel and team performance because it created disruption without requiring the defense to sacrifice a coverage player. Additional rushers remained a viable complementary mechanism, but their usage did not substitute consistently for four-man efficiency.

Pressure production was a room-level capability. The quality of the leading rusher was positively associated with overall and four-man pressure, but so was the quality of the supporting rotation. Star + Depth produced the strongest observed structure, while Distributed Rotation supplied a viable non-star pathway.

Resource quantity and concentration were weak substitutes for demonstrated player quality. Model fit improved materially only after prior room quality was introduced, and additional total resource commitment remained uninformative once room quality and the full room-construction controls were known. Room quality also retained forward-looking information after previous pressure was adjusted for the persistent ability of the prior season's headliner and supporting cast, with the cleanest separation for four-man pressure.

Taken together, the findings do not identify one optimal spending level or allocation structure. They indicate that the relevant organizational output is the quality structure produced by those resources and the extent to which that structure can generate pressure with four rushers.

9.2 Roster-construction implications

The results narrow the conventional choice between acquiring a star and building depth. Those approaches are not mutually exclusive, and neither label is sufficiently informative on its own.

A premium pass rusher can raise the ceiling of a defensive front, but the value of that player depends partly on the production available elsewhere in the room. Complementary pressure can be supplied by an interior creator, an above-average second edge defender, several rotational players, or developing contributors on rookie contracts. The requirement is credible production beyond the headliner, not a prescribed method of purchasing it.

This changes how a premium extension should be evaluated. The decision is not limited to whether the individual player is worth the contract in isolation. It also determines how much flexibility remains to preserve the rest of the room. A star-centered structure becomes fragile when the contract is accompanied by insufficient support, limited succession planning, or heavy dependence on the star's availability.

More broadly, the conditional resource result changes the interpretation of a room that has already demonstrated quality. Additional spending should not be assumed to produce a linear increase in pressure simply because it is directed toward the defensive front. Once a credible level of creation is established, the next contract or draft investment is better framed as a choice among preserving proven quality, replacing a vulnerable component, adding redundancy, or deploying the scarce resource elsewhere on the roster. This does not make

investment unnecessary; it makes the quality preserved or created by that investment the relevant intermediate outcome.

Distributed construction presents a different solution. Rather than concentrating production in one player, the team relies on several credible rushers. Distribution should not be confused with depth: spreading resources is valuable only when those players become effective contributors.

The practical implication is that resource decisions should begin with diagnosis. Teams should determine whether the current constraint is top-end creation, complementary quality, continuity, or developmental uncertainty. The next contract, draft pick, or acquisition should address that limiting feature rather than conform to a preferred allocation shape. Table 3 summarizes the observed construction pathways, their requirements, and their primary risks.

Table 3. Defensive-Front Construction Pathways, Requirements, and Primary Risks

Observed structure	Talent structure	What must be true	Pressure pathway	Primary risk	Evidentiary interpretation
Star + Depth	Elite primary creator with credible complementary quality	The headliner remains productive and the supporting rotation clears a meaningful quality floor.	High-end creation plus multiple secondary sources of four-man pressure.	Injury, decline, or departure of the star; support erosion if resources become too concentrated.	Strongest observed pressure distribution. Supports pairing top-end creation with credible support; does not establish a universally optimal structure.
Distributed Rotation	No qualifying creation star, but several credible rushers and strong collective support	Evaluation and development must produce multiple above-average contributors simultaneously.	Pressure generated collectively rather than through one dominant creator.	Several personnel decisions must succeed; loss of depth can reduce the structure quickly.	Viable non-star pathway. Demonstrates that strong pressure production need not depend on one dominant player.
Heliocentric Star	Qualifying creation star with weaker complementary quality	The headliner must supply an unusually large share of the room's disruption.	Pressure is disproportionately dependent on the leading creator.	Highest exposure to the star's health, decline, protection attention, or absence; limited secondary creation.	Higher ceiling than thin rooms but less complete than Star + Depth. A star did not consistently substitute for support quality.
Thin / Mixed	Neither sufficient top-end creation nor strong collective support	Meaningful improvement must come from acquisition, development, role change, or internal progression.	No consistently demonstrated room-level pressure pathway.	Low pressure floor and multiple unresolved roster needs.	Diagnostic baseline rather than a recommended strategy. Produced the weakest observed pressure distribution.

Notes: Archetypes summarize observed defensive-front talent structures using the study's predefined headliner- and support-quality rules. They are descriptive rather than estimated optimal strategies. Star + Depth and Heliocentric Star contain a qualifying creation star and differ primarily in supporting-cast quality; Distributed Rotation contains no qualifying star but several credible creators; Thin / Mixed meets none of the other definitions. The table summarizes empirical patterns and associated roster risks; it does not prescribe a salary-cap share, draft allocation, player count, or resource-concentration target.

Table 3. Defensive-front construction pathways, requirements, and primary risks.

9.3 Multiple Pathways, Different Failure Modes

The empirical results do not identify one universally superior method of defensive-front construction. Strong pressure production appeared under different talent and resource structures, while resource quantity and concentration alone provided limited information about subsequent pressure. The relevant organizational question is therefore not which allocation shape is inherently optimal, but how a chosen structure is expected to convert resources into demonstrated pass-rush quality. Four recognizable construction approaches illustrate that distinction: a premium pipeline, a star duo, heliocentric construction, and a distributed sum-of-the-parts model. Each provides a plausible route to a productive four-man rush, and each creates a characteristic failure mode. The paired cases below are illustrative applications of the empirical framework rather than additional statistical tests.

Premium Pipeline — Philadelphia and Cincinnati

The premium-pipeline model repeatedly directs high-value acquisition and retention resources toward the defensive front. Its advantage is succession. Philadelphia illustrates the upside: repeated premium investment can replenish the room before previous production disappears, allowing top-end and complementary quality to coexist across roster cycles. The relevant feature for this study is not simply high investment, but successful conversion of repeated resource commitments into demonstrated creation throughout the room.

Cincinnati illustrates the corresponding pitfall. Continued investment around an established high-end creator does not guarantee that complementary players develop into independent sources of pressure. Draft capital measures organizational commitment at the time of acquisition, not the quality ultimately produced. A premium pipeline therefore reduces replenishment risk only when repeated investments become productive players; otherwise, substantial resource commitment can coexist with continued dependence on the incumbent star.

Star Duo — 2025 Houston and the 2024 Los Angeles Chargers

A star-duo structure attempts to place two premium creators at the center of the four-man rush. Houston illustrates the upside: two high-end creators can establish the pressure ceiling while allowing the remainder of the room to be built around complementary roles rather than asking every rotational player to become a difference-maker. When both players generate pressure without routinely requiring additional rushers, the structure can create a particularly strong four-man pathway.

The 2024 Los Angeles Chargers illustrate the central pitfall. A roster can allocate resources as though it possesses a premium duo without receiving premium-duo production. Concentrating commitment in two players does not itself create two high-level pressure producers. A star-duo resource structure is therefore not equivalent to having a star duo. The model succeeds only when both major investments continue to provide the creation expected from their resource commitment, and it becomes much less efficient when one or both fall short.

Heliocentrism — 2025 Cleveland and the 2022 Los Angeles Rams

Heliocentric construction accepts unusually high dependence on one exceptional creator. Cleveland illustrates the upside. A transcendent headliner can raise the ceiling of the entire front and reduce the amount of independent creation required from complementary players. In that setting, concentration is not necessarily evidence of poor construction; it can reflect a rational attempt to exploit an unusually scarce player.

The 2022 Los Angeles Rams illustrate the corresponding exposure. When a large share of the room's pressure ceiling depends on one player, injury, decline, or erosion of the surrounding cast can affect the entire structure disproportionately. The case is not evidence that one player's availability alone caused the team's change in pressure performance. Rather, it illustrates the dependency inherent in the model: heliocentric rooms have less margin for deterioration at their central node. The same feature that creates the upside also concentrates replacement difficulty and performance risk.

Sum of the Parts — 2025 Seattle and 2025 Baltimore

The sum-of-the-parts model takes the opposite approach. Instead of requiring one or two dominant creators, pressure is generated collectively through several credible contributors. Seattle illustrates the upside. Multiple players can provide enough independent creation that the aggregate front generates strong pressure without requiring any single rusher to dominate the room's production. This pathway is consistent with the Layer 2 finding that complementary creation can itself become the foundation of an effective room rather than merely serving as support for a star.

The 2025 Baltimore case illustrates the limitation. A collection of individually plausible contributors does not necessarily produce an effective collective rush. Distributed construction requires several evaluation, development, deployment, and retention decisions to succeed simultaneously. When enough components fall below expectation, the absence of a singular creator becomes more consequential. The model exchanges concentrated dependence for distributed execution risk.

Taken together, these paired cases reinforce the paper's central construction implication: there is no infallible defensive-front architecture. Premium replenishment depends on repeated evaluation and development success. A star duo requires both premium investments to perform like premium creators. Heliocentric construction accepts unusually high dependence on one player. A sum-of-the-parts approach requires several personnel decisions to succeed together. Resource allocation therefore helps determine where an organization chooses to carry risk rather than eliminating that risk. The objective is not to identify one structure that is universally optimal, but to understand the dependency created by the chosen model and determine whether the resulting room has converted its resources into enough demonstrated quality to sustain a credible four-man pressure pathway.

9.4 Construction risk and organizational interpretation

The viable structures carry different forms of risk. Heliocentric rooms are exposed directly to the health, decline, or departure of the leading player. Distributed rooms reduce direct reliance on one creator but require several personnel decisions to succeed simultaneously.

Neither risk profile was independently resolved by resource concentration. The interaction analysis provided no evidence that concentration consistently strengthened or weakened the relationship between room quality and pressure. Allocation shape describes the location of organizational exposure, not the quality of the resulting room.

Continuity should be interpreted similarly. Returning rush share provided secondary positive signal, consistent with preserving known production and reducing replacement uncertainty. It does not follow that continuity is desirable regardless of quality.

The weak estimates for unproven resource share and alignment also warrant restraint. These measures describe relevant organizational conditions, but the current evidence does not support strong universal rules about either.

9.5 Contribution and interpretation boundaries

The primary contribution is the integration of team pressure, player-level room structure, and resource allocation within a single empirical framework. Layer 1 identifies pressure as a meaningful defensive capability. Layer 2 shows that this capability reflects both top-end and complementary player quality. Layer 3 shows that spending quantity and concentration do not reliably describe whether that quality has been produced and, once demonstrated room quality is known, additional total resource commitment provides little incremental information about subsequent pressure.

The findings support a distinction between resource commitment and resource conversion. Cap space and draft capital create access to talent. They do not guarantee that acquired players will develop, remain healthy, fit their roles, or collectively generate pressure. Once productive quality is demonstrated, resource allocation becomes a retention and opportunity-cost problem as much as an acquisition problem.

The study does not estimate a causal return to cap spending, draft capital, continuity, or concentration. The results should not be interpreted as establishing an optimal cap share, player count, or concentration threshold. They provide a framework for evaluating whether a chosen strategy has produced sufficient creation quality and a credible four-man pressure pathway.

10. Limitations and Future Research

10.1 Scheme, protection, and pressure attribution

The primary football limitation is the absence of reliable play-level tags for stunts, twists, simulated pressures, and creepers. The data identify how many defenders rushed, but not the complete design through which pressure was created.

Four-man pressure should therefore be interpreted as a defensive capability rather than a pure measure of four individual rushers winning independently. A four-man pressure may result from a conventional rush, a simulated pressure, or a coordinated exchange that manipulates protection responsibilities.

Player attribution is limited for the same reason. On a stunt, the defender receiving the pressure credit may not be the player who created the protection conflict. Protection attention is also unobserved. Double teams, chips, slides, and backfield help may reduce a headliner's recorded pressure while improving opportunities for teammates. Tracking and interaction-level research shows why these assignments matter for trench evaluation (Hong et al. 2024; Pipping-Gamon et al. 2026).

Coverage structure, quarterback mobility, and time to throw may also affect whether pressure develops. The results describe observed pressure production, not a fully isolated pass-rush mechanism.

10.2 Player and resource measurement

The player-quality measures are derived from observed opportunities and remain sensitive to deployment, alignment, matchup, and role. Players are not assigned randomly to pass-rush situations.

The room definition includes edge and defensive interior defenders. Pressure from linebackers, defensive backs, and hybrid players appears in team pressure rates but not in the player-based room-quality measure. Injuries are represented indirectly through opportunity and continuity rather than through a separate health model.

Players without a valid prior estimate receive a neutral 50th-percentile value, while their resource share is tracked as unproven. This avoids treating missing information as poor performance, but combines rookies, injured players, and low-opportunity veterans within one category.

The resource index approximates organizational commitment rather than full economic cost. Cap charges reflect accounting structure, while draft-value measures approximate opportunity cost. Neither captures private evaluations, development spending, or team-specific alternatives.

10.3 Identification and generalizability

The analysis is observational. Teams allocate resources in response to expected player quality, roster need, scheme, contract availability, and competitive timelines. The estimates should not be interpreted as causal returns to spending, continuity, or concentration.

Layer 3 contains 82 qualifying team-seasons from 2023-2025. This supports broad comparisons but limits precision for interactions, nonlinearities, and narrow allocation categories. Coaching, development, health, and organizational quality may influence both room construction and pressure. The conditional resource-by-quality interactions were imprecise and should not be interpreted as identifying a specific ability threshold above which additional spending ceases to matter.

The 2025 holdout provides retrospective temporal validation rather than a fully prospective test. The broader framework was developed with knowledge of the complete study period, and the four-man models overpredicted the 2025 pressure environment. The validation is therefore stronger as evidence of relative differentiation than exact calibration.

10.4 Future research

The highest-priority extension is a play-level decomposition of pressure design. Reliable tags for stunts, twists, simulated pressures, creepers, rush lanes, and post-snap role exchanges would separate individual wins from coordinated pressure creation.

Protection data would provide the next major improvement. Double-team rates, chips, slides, blocker assignments, and protection declarations would clarify the indirect value created by elite rushers and the extent to which support production reflects favorable attention.

Future models should also incorporate alignment-specific quality, hybrid rushers, projected opportunity, age, health, and role stability. Additional seasons would permit rolling temporal validation and a genuinely prospective test of the frozen framework.

The present study identifies demonstrated room quality and four-man pressure capability as more informative than resource shape alone. It does not fully determine how much of that capability is produced by individual talent, coordinated design, protection response, or organizational development.

11. Conclusion

NFL teams face a constrained roster problem: consistent pressure is valuable, but the players most capable of creating it are scarce and expensive. The evidence here does not support a universal choice between star power and depth. It supports a narrower objective: build a room with enough demonstrated quality to sustain pressure with four rushers.

Teams reached that objective through more than one structure. A high-end creator can anchor the front, but the strongest star-led rooms also maintained credible complementary production. Distributed rotations offered a viable alternative when several rushers collectively supplied sufficient quality. Star-centered rooms are more exposed to one player's availability and decline, while distributed rooms depend on several evaluation, development, and retention decisions succeeding together.

Resource allocation alone did not resolve that choice. Total investment and concentration were weak direct predictors of pressure, and once demonstrated room quality and the full room-construction controls were known, additional total resource commitment provided little incremental information about subsequent pressure. Cap space and draft capital remain necessary inputs, but their value depends on whether they convert into productive players, retained quality, and a functional four-man rush. Spending shape should therefore be evaluated as a means rather than an outcome.

The practical decision rule is straightforward. Before committing the next contract or draft selection, a team should assess whether the projected room has a credible four-man pressure path, whether meaningful creation exists beyond its best player, how much established production will return, and which failure mode the chosen structure creates.

This study does not identify an optimal cap share, player count, or acquisition strategy. It provides a framework for determining whether scarce resources have produced the player quality, complementary creation, and four-man pressure capability the defense requires.

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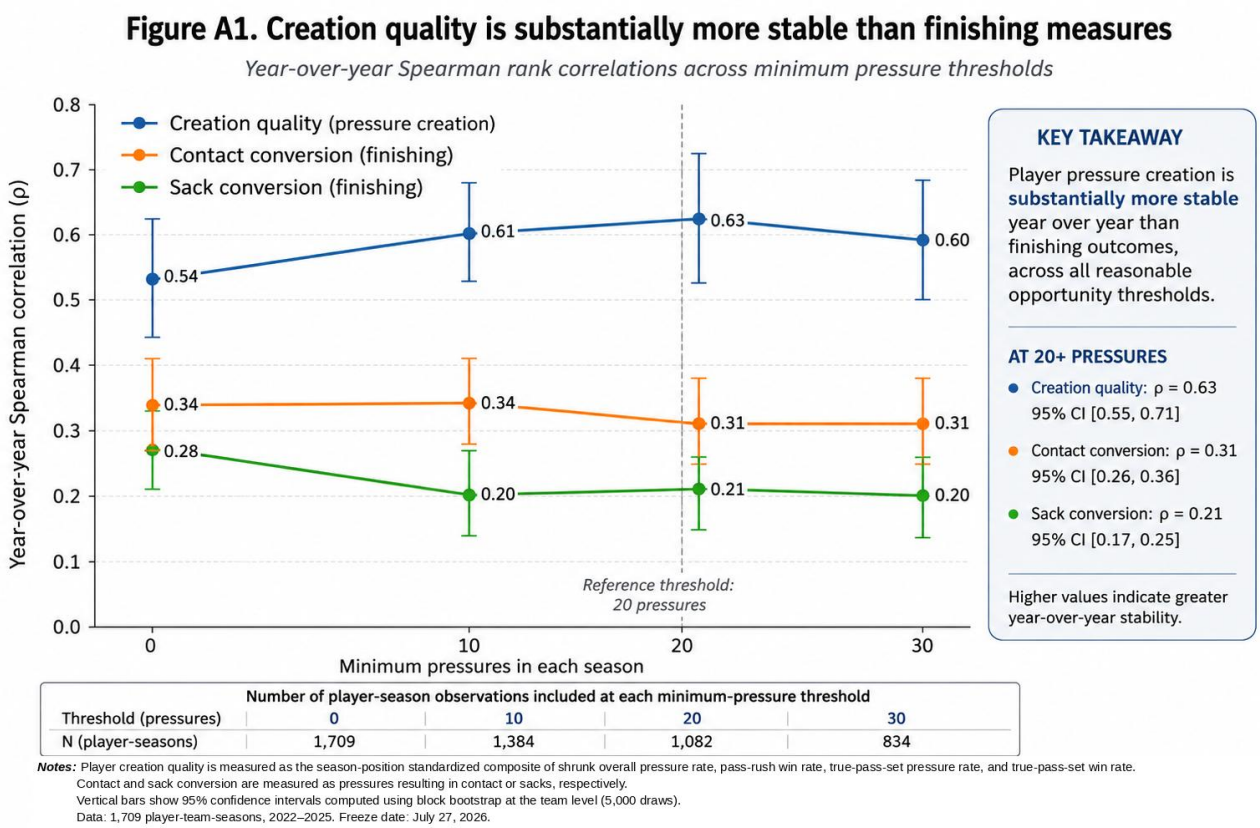
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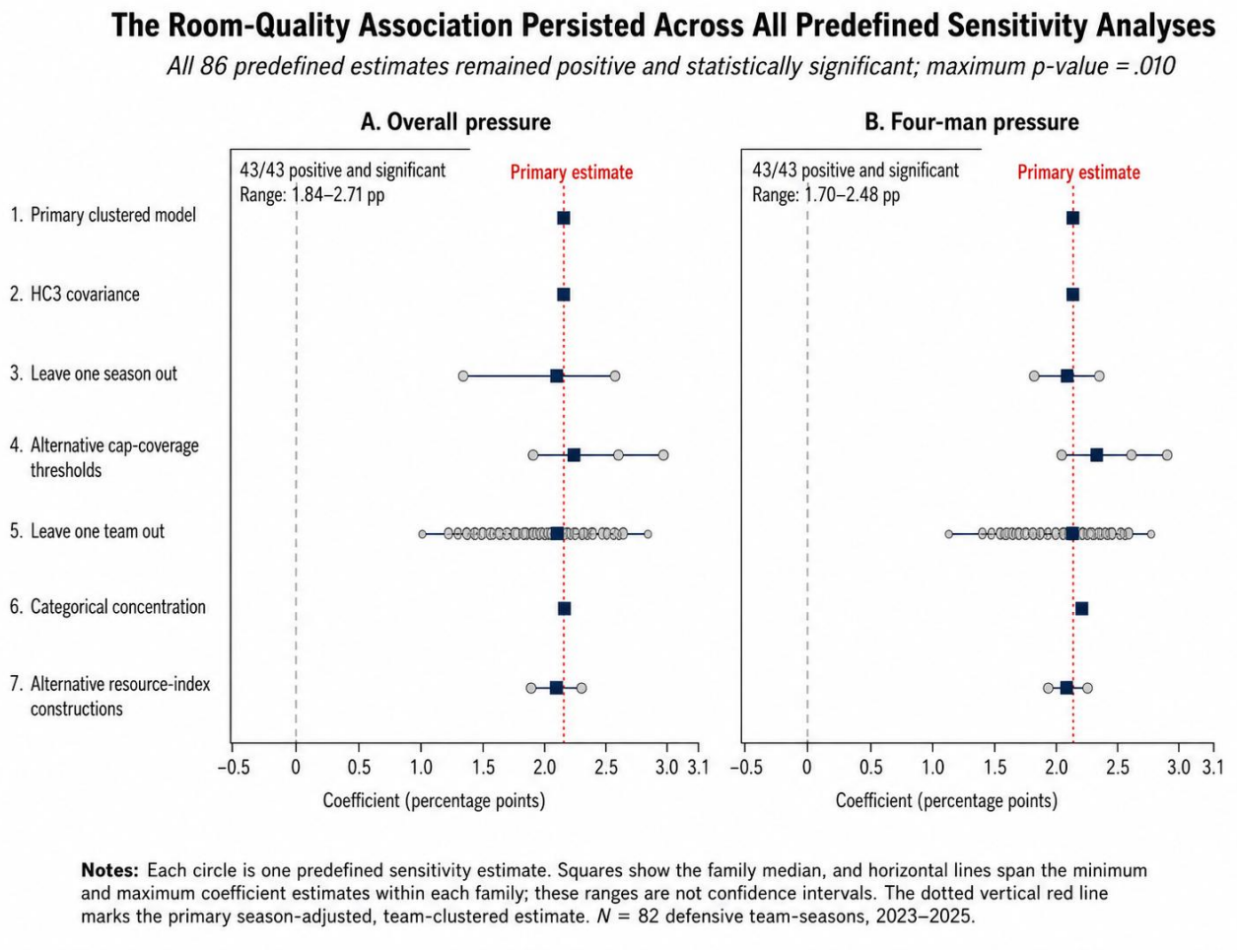
Appendix Exhibits



Number of player-season observations included at each minimum-pressure threshold				
Threshold (pressures)	0	10	20	30
N (player-seasons)	1,709	1,384	1,082	834

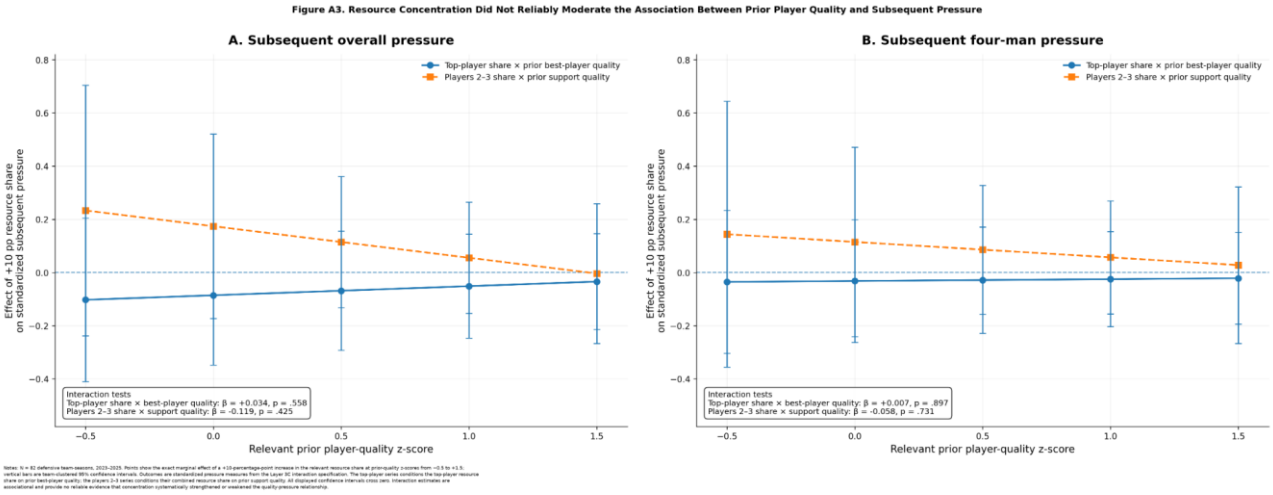
Notes: Player creation quality is measured as the season-position standardized composite of shrunk overall pressure rate, pass-rush win rate, true-pass-set pressure rate, and true-pass-set win rate. Contact and sack conversion are measured as pressures resulting in contact or sacks, respectively. Vertical bars show 95% confidence intervals computed using block bootstrap at the team level (5,000 draws). Data: 1,709 player-team-seasons, 2022–2025. Freeze date: July 27, 2026.

Appendix Figure A1. Year-over-year stability of creation quality and finishing measures.



Notes: Each circle is one predefined sensitivity estimate. Squares show the family median, and horizontal lines span the minimum and maximum coefficient estimates within each family; these ranges are not confidence intervals. The dotted vertical red line marks the primary season-adjusted, team-clustered estimate. $N = 82$ defensive team-seasons, 2023–2025.

Appendix Figure A2. Prior room-quality sensitivity analysis across predefined specifications.



Appendix Figure A3. Resource concentration interactions with prior player quality and subsequent pressure.

Appendix Table A4. Additional Layer 3 Sensitivity Tests and FTN-PFF Concordance

Peer-review analyses assessing conditional resource effects, player-adjusted persistence, and cross-provider measurement agreement.

Panel A. Conditional total-resource association after demonstrated room quality

Outcome	Total resource index	95% CI / p	Prior room quality (+10 pct pts)	95% CI / p	Resource × quality
Overall pressure	-0.39 pp	[-1.06, 0.28]; p = .255	+2.19 pp	[1.18, 3.20]; p < .001	-0.45; p = .276
Four-man pressure	-0.51 pp	[-1.12, 0.10]; p = .100	+2.04 pp	[1.03, 3.05]; p < .001	-0.38; p = .300

Panel B. Player-adjusted prior-pressure persistence

Outcome / specification	Prior room quality	95% CI / p	Residual prior pressure	95% CI / p	Adjusted R^2
Overall: quality + residual	+2.02 pp	[0.99, 3.05]; p < .001	+0.294	[0.068, 0.521]; p = .011	0.228
Overall: full room + residual	+2.18 pp	[1.17, 3.19]; p < .001	+0.244	[0.016, 0.472]; p = .036	0.256
Four-man: quality + residual	+1.84 pp	[0.80, 2.88]; p < .001	+0.106	[-0.061, 0.273]; p = .215	0.162
Four-man: full room + residual	+2.06 pp	[1.05, 3.07]; p < .001	+0.092	[-0.100, 0.284]; p = .349	0.242

Panel C. FTN-PFF defense-season concordance

Comparison	Scale	Pearson r	Spearman rho	N / significance
PFF ED/DI room pressure vs FTN overall pressure	Within-season z-score	0.724	0.694	128; p < .001
PFF ED/DI room pressure vs FTN four-man pressure	Within-season z-score	0.718	0.689	128; p < .001

Notes: Layer 3 sensitivity models use N = 82 defense-front team-seasons from 2023-2025, season fixed effects, and team-clustered standard errors. Room quality is scaled per 10 percentile points; total resource index is standardized. Residual prior pressure is observed prior-season pressure minus leave-one-team-out expected pressure from same-season headliner and support quality. FTN measures binary team pressured-dropback outcomes, whereas PFF supplies player-attributed ED/DI pressure production; concordance rather than numerical equality is the relevant reconciliation test. The conditional resource interaction does not establish a spending threshold. All inputs remain frozen to the July 27, 2026 project snapshot.

	A	B	C	D	E	F	G	H
1	**Appendix Table A1. Full Layer 1 Model Results**							
2	*Pressure–defense relationships, probability reference points, pressure pathways, and secondary outcomes*							
3								
4	**Specification**	**Outcome**	**Predictor / estimand**	**Estimate**	**95% CI**	**p-value**	**N**	**Model fit / diagnostic**
5	**Panel A. Primary pressure–EPA model**							
6	Pooled OLS	EPA quality	Within-season pressure percentile	0.117	—	< .001	128	$R^2 = 0.261$
7	5,000-draw bootstrap	EPA quality	Interquartile-range pressure shift	0.0587	[0.039, 0.077]	—	128	Bootstrap effect estimate
8								
9	**Panel B. Probability models and nonlinear check**							
10	Logistic GLM	Average-or-better EPA tier	50% fitted-probability crossing	33rd percentile	[13th, 47th]	—	128	Viability reference; not a hard threshold
11	Logistic GLM	Good-or-elite EPA tier	50% fitted-probability crossing	66th percentile	[50th, 86th]	—	128	Viability reference; not a hard threshold
12	Linear vs. spline	EPA quality	Nonlinear fit comparison	No material improvement	—	—	128	Linear relationship retained

Appendix Table A1. Full Layer 1 Model Results (continued)

Pressure–defense relationships, probability reference points, pressure pathways, and secondary outcomes

Specification	Outcome	Predictor / estimand	Estimate	95% CI	p-value	N	Model fit / diagnostic
Panel C. Joint pressure-pathway model							
Joint pathway OLS	EPA quality	Four-man pressure efficiency	0.787	—	< .001	128	Full R^2 = 0.293; ΔR^2 if removed = 0.095
Joint pathway OLS	EPA quality	Five-plus pressure efficiency	0.31	—	0.01	128	Full R^2 = 0.293; ΔR^2 if removed = 0.036
Joint pathway OLS	EPA quality	Five-plus usage	0.05	—	n.s.	128	Frequency not independently distinguishable from zero
Panel D. Secondary defensive outcomes							
Secondary pathway OLS	Success-rate prevention	Four-man pressure efficiency	0.186	—	0.021	128	Directionally consistent with primary pathway result
Secondary pathway OLS	Success-rate prevention	Five-plus pressure efficiency	0.128	—	0.057	128	Positive estimate; weaker evidence
Secondary pathway OLS	20+ explosive-pass prevention	Four-man and five-plus pathways	Not reliably associated	—	—	128	No stable pathway relationship

Notes

1 Unit of analysis is the defense-season. The Layer 1 sample contains 128 defense-seasons from 2022–2025 and 80,719 eligible dropbacks.

2 EPA quality is EPA allowed per eligible dropback, sign-oriented so that higher values indicate better defense. The primary pressure predictor is within-season pressure percentile.

3 The pooled models are same-season associations and should not be interpreted causally. Season adjustment is used to account for changes in the league pressure environment.

4 The joint pathway model includes four-man pressure efficiency, five-plus pressure efficiency, and five-plus usage simultaneously. ΔR^2 reports the reduction in model R^2 when the named efficiency term is removed.

5 The 33rd- and 66th-percentile crossing points are fitted-probability reference points with bootstrap uncertainty, not necessary or sufficient pressure thresholds.

6 Explosive-pass prevention refers to prevention of pass plays gaining at least 20 yards. An em dash indicates that the statistic is not reported in the frozen Layer 1 manuscript outputs, not that its value is zero.

7 Source basis: frozen Layer 1 outputs and the stitched August 2026 manuscript; all project data were frozen July 27, 2026.

	A	B	C	D	E	F	G	H
1	**Appendix Table A2. Full Layer 2 Continuous-Model Results**							
2	*Simultaneous headliner- and support-quality relationships with overall and four-man team pressure*							
3								
4	**Panel**	**Outcome**	**Predictor**	**Effect scale**	**Estimate**	**95% CI**	**p-value**	**N**
5	Panel A. Overall pressure	Overall team pressure rate	Headliner quality	Percentage-point change per 1-SD increase	0.96	—	—	128
6	Panel A. Overall pressure	Overall team pressure rate	Support quality	Percentage-point change per 1-SD increase	1.86	—	—	128
7								
8	Panel B. Four-man pressure	Four-man pressure rate	Headliner quality	Percentage-point change per 1-SD increase	0.9	—	—	128
9	Panel B. Four-man pressure	Four-man pressure rate	Support quality	Percentage-point change per 1-SD increase	1.9	—	—	128
10								
11	**Notes**							
12	*1 Unit of analysis is the defense-season. Layer 2 contains all 128 defense-seasons from 2022–2025.*							
13	*2 Headliner quality represents the creation quality of the room's highest creation-score player. Support quality is the opportunity-weighted creation quality of the remaining qualifying ED/DI players.*							
14	*3 Headliner and support quality are entered simultaneously in season-adjusted ordinary least squares models. Overall and four-man pressure are modeled separately.*							
15	*4 Estimates are expressed as percentage-point changes in the relevant pressure rate for a one-standard-deviation increase in the named quality measure.*							
16	*5 The models are same-season associations and should not be interpreted causally. Continuous measures carry the primary inferential weight; archetype comparisons are descriptive summaries.*							
17	*6 An em dash indicates that the statistic is not exposed in the current frozen stitched-manuscript outputs available for appendix reconstruction. It is not a zero and has not been imputed from an earlier standardized specification.*							
18	*7 Source basis: frozen Layer 2 analysis and the August 2026 stitched manuscript; project data were frozen July 27, 2026.*							

Appendix Table A3. Frozen Room-Archetype Definitions and Structural Profiles

Rule-based Layer 2 defensive-front talent structures, 2022–2025

Panel A. Frozen archetype definitions

Archetype	N	Frozen operational definition
Star + Depth	21	Creation Star (creation score >= 0.95) with supporting-cast quality at or above the frozen 60th team-season percentile.
Distributed Rotation	31	No Creation Star; strong supporting-cast quality; and at least three above-average ED/DI rushers with 100+ pass-rush snaps.
Heliocentric Star	21	Creation Star (creation score >= 0.95) with supporting-cast quality below the frozen 60th team-season percentile.
Thin / Mixed	55	All remaining rooms: no qualifying star with insufficient distributed quality, or otherwise weak/mixed construction.

Appendix Table A3. Frozen Room-Archetype Definitions and Structural Profiles (continued)

Rule-based Layer 2 defensive-front talent structures, 2022–2025

Panel B. Structural profiles (medians)

Archetype	N	Headliner creation score	Support quality (z)	Support percentile	Above-average rushers (100+ PR snaps)	Opportunity HHI	Pressure HHI	Top-1 pressure share	Effective rushers	Top-two quality (z)	Third-plus quality (z)
Star + Depth	21	0.972	0.518	87.5%	6	0.122	0.155	28.0%	8.23	2.073	0.234
Distributed Rotation	31	0.93	0.258	78.1%	5	0.115	0.137	20.2%	8.66	1.244	0.046
Heliocentric Star	21	0.976	-0.361	31.2%	3	0.128	0.187	29.6%	7.81	1.818	-0.628
Thin / Mixed	55	0.889	-0.361	31.2%	3	0.124	0.153	23.7%	8.04	0.835	-0.561

Notes

1. *Unit of analysis is the defense-season. Layer 2 contains 128 defense-seasons from 2022–2025.*
2. *The headliner is the room's highest creation-score player. Creation Score is the geometric mean of creation percentile and volume percentile; a Creation Star has a score of at least 0.95.*
3. *Support quality is the pass-rush-snap-weighted mean creation quality of the remaining qualifying PFF edge and defensive-interior players. The 60th-percentile line is an archetype boundary, not a universal roster threshold.*
4. *Above-average rushers counts qualifying ED/DI rushers above the relevant creation-quality benchmark with at least 100 pass-rush snaps.*
5. *Opportunity HHI is the concentration of pass-rush opportunity (snap HHI); pressure HHI is the concentration of player-attributed pressures. Higher HHI indicates greater concentration.*
6. *Effective rushers equals 1 / opportunity HHI. The displayed median opportunity HHI is recovered from the median effective-rusher value; this is exact for these odd-sized archetype groups because the reciprocal is monotonic.*
7. *Top-1 pressure share is the share of room-attributed pressures produced by the leading pressure producer. Top-two and third-plus quality summarize creation quality within those portions of the room.*
8. *The archetypes are transparent, rule-based descriptive summaries of realized talent structure. They are not latent classes, spending strategies, or measures of organizational intent; continuous room-quality measures carry the primary inferential weight.*
9. *Source basis: frozen Layer 2 final findings, room-construction analysis, and team-season room master. All project data were frozen July 27, 2026.*

	A	B	C	D	E	F
1	**Appendix Table A5. Robustness and Model-Specification Registry**					
2	***Sensitivity checks used to determine which Pressure Creation claims survive alternative specifications, samples, outcomes, and inference choices***					
3						
4	**Layer / domain**	**Robustness dimension**	**Primary / reference specification**	**Alternative check**	**Observed result**	**Claim disposition**
5	**Layer 1**	Temporal stability	Season-adjusted pressure–EPA model	Leave-one-season-out estimation	Pressure–EPA relationship remained positive in every leave-one-season-out specification.	**Retained**
6	**Layer 1**	Functional form	Linear pressure–EPA relationship	Parsimonious nonlinear / spline comparison	Spline specification produced no material improvement over the linear relationship.	**Retained with boundary: no hard pressure threshold inferred**
7	**Layer 1**	Outcome substitution	EPA quality	Success-rate prevention	Pressure remained positively associated with success-rate prevention.	**Retained**
8	**Layer 1**	Outcome boundary / placebo	EPA and success-rate outcomes	Prevention of pass plays gaining at least 20 yards	Pressure pathways were not reliably associated with 20+ explosive-pass prevention.	**Boundary retained: pressure is not a complete pass-defense explanation**
9	**Layer 1**	Rush-pathway decomposition	Overall pressure	Four-man efficiency, five-plus efficiency, and five-plus usage entered jointly	Four-man and five-plus pressure efficiency were positively associated with defensive quality; five-plus usage was not independently informative.	**Retained with tactical boundary**
10						
11	**Layer 2**	Continuous vs. categorical representation	Rule-based room archetypes	Season-adjusted continuous headliner- and support-quality models	Headliner and support results remained directionally consistent when represented continuously rather than through archetypes.	**Retained; continuous measures carry primary inferential weight**
12	**Layer 2**	Archetype interpretation	Frozen Star + Depth / Distributed Rotation / Heliocentric Star / Thin-Mixed labels	Claim-discipline review against continuous models	Observed pathway contrasts remain useful descriptively, but exact thresholds and category contrasts are not treated as universal latent classes.	**Descriptive only**

	A	B	C	D	E	F
1	**Appendix Table A5. Robustness and Model-Specification Registry (continued)**					
2	***Sensitivity checks used to determine which Pressure Creation claims survive alternative specifications, samples, outcomes, and inference choices***					
3						
4	**Layer / domain**	**Robustness dimension**	**Primary / reference specification**	**Alternative check**	**Observed result**	**Claim disposition**
5	**Layer 3**	Uncertainty estimator	Season indicators with team-clustered standard errors	HC3 heteroskedasticity-robust standard errors	Prior room-quality estimates remained positive under both uncertainty estimators.	**Retained**
6	**Layer 3**	Draft-value construction	Primary strict Fitzgerald-Spielberger / OTC four-year annualized value	Alternative draft-value curves	Prior room-quality estimates remained positive across alternative draft-value constructions.	**Retained**
7	**Layer 3**	Cap-coverage sample	Primary sample: >=80% opportunity-weighted cap coverage (N=82)	Alternative cap-coverage thresholds	Prior room-quality estimates remained positive across alternative coverage samples.	**Retained**
8	**Layer 3**	Temporal influence	Full 2023–2025 lagged panel	Leave-one-season-out estimation	No single season determined the positive prior room-quality result.	**Retained**
9	**Layer 3**	Team influence	Full qualifying lagged panel	Leave-one-team-out estimation	No single team determined the positive prior room-quality result.	**Retained**
10	**Layer 3**	Influential observations	Primary nested models	Influence diagnostics / unusual team-season exclusions	Several unusual team-seasons were identified, but the room-quality conclusion did not depend on a single observation.	**Retained**
11	**Layer 3**	Allocation-shape decomposition	Continuous resource commitment and HHI	Frozen Single-Player Bet / Two-Player Core / Three-Player Core / Distributed Investment indicators	Joint Wald tests for allocation-shape indicators were not significant: overall pressure p=.222; four-man pressure p=.336.	**Direct allocation-shape claim not retained**
12	**Layer 3**	Direct resource measure	Total resource commitment	Resource HHI, leading-player share, and combined shares assigned to players ranked second and third	Across the four direct resource measures and two pressure outcomes, all eight 95% confidence intervals crossed zero.	**Direct resource-quantity / concentration claim not retained**
13	**Layer 3**	Quality × concentration interaction	Additive prior room quality + resource HHI model	Prior room quality × resource HHI interaction	Interaction was small and imprecise: overall beta=-0.34, p=.409; four-man beta=-0.38, p=.392.	**No moderation claim retained**

	A	B	C	D	E	F
1	**Appendix Table A5. Robustness and Model-Specification Registry (continued)**					
2	***Sensitivity checks used to determine which Pressure Creation claims survive alternative specifications, samples, outcomes, and inference choices***					
3						
4	**Layer / domain**	**Robustness dimension**	**Primary / reference specification**	**Alternative check**	**Observed result**	**Claim disposition**
5	**Layer 3**	Continuity	Nested room-construction model	Returning-rush-share term	Returning rush share provided secondary positive signal, but estimates were smaller and less stable than the room-quality effect.	**Secondary**
6	**Layer 3**	Developmental uncertainty	Nested room-construction model	Unproven resource-share term	Unproven resource share did not produce a stable independent effect.	**Unresolved; no directional interpretation**
7	**Layer 3**	Resource-quality alignment	Nested room-construction model	Resource-weighted quality alignment term	Alignment was positive but imprecise.	**Exploratory**
8	**Layer 3**	Persistence control	Prior team pressure alone	Prior player-based room quality alone and jointly with prior pressure	Adding room quality improved explanatory fit for both outcomes; the distinction was clearest for four-man pressure, where room quality remained significant and previous team four-man pressure did not.	**Retained, strongest for four-man pressure**
9	**Layer 3**	Retrospective temporal validation	2023–2024 training sample; 2025 qualifying holdout	Prior-pressure baseline vs. room-quality and complete-room models	Room quality and the complete room model outperformed the prior-pressure baseline for four-man pressure. Overall-pressure gains were smaller; four-man models overpredicted the 2025 league environment.	**Predictive differentiation retained; exact calibration qualified**
10						
11	**Evidence hierarchy**	Claim-survival synthesis	Primary manuscript claims	Integrated review across model, sample, temporal, measurement, and outcome checks	Demonstrated player / room quality is supported. Direct resource quantity and concentration are unsupported as general predictors. Continuity is secondary, alignment exploratory, and unproven resource share is treated as uncertainty rather than low quality.	**Frozen evidence hierarchy**
12						
13	**Notes**					
14	***1*** *This registry records robustness checks and their manuscript-level interpretation; it is not a substitute for the full coefficient tables or Appendix Figures A1–A3.*					
15	***2*** *Layer 3 primary inference uses season indicators and team-clustered standard errors because organizations contribute repeated team-seasons. HC3 estimates are used as an alternative uncertainty check.*					
16	***3*** *The primary Layer 3 sample contains 82 defense-front team-seasons from 2023–2025 meeting the predefined 80% opportunity-weighted cap-coverage threshold.*					
17	***4*** *The primary draft input is the strict Fitzgerald-Spielberger / Over The Cap value, annualized over four seasons and credited to homegrown players in seasons zero through three of the rookie-contract window.*					
18	***5*** *A retained claim means its direction and substantive interpretation survived the relevant checks. It does not imply causal identification or universal applicability.*					
19	***6*** *Null or imprecise estimates are interpreted as a lack of reliable evidence in the observed sample, not proof that the underlying mechanism can never matter.*					
20	***7*** *Source basis: frozen Layer 1–3 outputs and the stitched August 2026 manuscript. All project data were frozen July 27, 2026.*					

Appendix Table A6. Reproducibility and Analytic-Panel Audit

Frozen data lineage, sample dimensions, temporal separation, and integrity checks for the Pressure Creation analysis

Panel A. Source and measurement provenance

Component	Primary source	Seasons	Analytical role	Frozen transformation / rule	Audit status	Reproducibility note
Team pressure and defensive outcomes	FTN play-by-play / team-defense data	2022–2025	Eligible dropbacks, pressured dropbacks, rusher-count pathways, EPA, success, explosive-pass outcomes	Eligible dropbacks include pass attempts, sacks, and QB scrambles; nullified plays removed; pressure aggregated as a binary team dropback event.	**Frozen**	FTN defines team-level pressure and outcomes throughout Layer 1 and downstream team-season outcomes.
Player pass-rush performance and room membership	Pro Football Focus	2022–2025	ED/DI opportunities, pressures, wins, true-pass-set metrics, sacks, hits, hurries, and position	Primary room restricted to PFF-classified edge and defensive-interior players; traded player-seasons segmented by observed team opportunity where required.	**Frozen**	PFF player pressure credits are not required to reconcile mechanically to FTN team-pressure events.
Salary-cap commitment	Spotrac	2022–2025	Player cap-hit inputs for Layer 3 resource measurement	Matched room cap hits divided by official season cap; opportunity coverage tracked; resource inputs standardized within season.	**Frozen**	Stable player identities, team abbreviations, trades, and cap matches were audited before final estimation.
Draft commitment	Strict Fitzgerald-Spielberger / Over The Cap draft value	2022–2025 inputs; Layer 3 outcomes 2023–2025	Homegrown draft-capital input to combined resource index	Draft value annualized over four seasons and credited in seasons 0–3 of the rookie-contract window.	**Frozen**	Alternative draft-value constructions retained for robustness rather than substituted into the primary panel.
Project extraction / accounting snapshot	All retained source and derived data	2022–2025	Locks the analytical information set used for final estimation	All project data frozen July 27, 2026.	**Frozen**	Later source updates are not incorporated into the submitted analytical panel unless the freeze is deliberately reopened.

Appendix Table A6. Reproducibility and Analytic-Panel Audit (continued)

Frozen data lineage, sample dimensions, temporal separation, and integrity checks for the Pressure Creation analysis

Panel B. Analytic sample dimensions

Analytic object	Unit of analysis	Study period	Potential / source universe	Final analytic N	Primary inclusion rule	Audit status
Layer 1 team panel	Defense-season	2022–2025	32 defenses × 4 seasons	128 defense-seasons; 80,719 eligible dropbacks	Valid defensive team-season with eligible FTN dropbacks and required pressure / outcome fields.	**Complete**
Layer 2 player-room master	PFF ED/DI player-team-season row	2022–2025	PFF edge and defensive-interior player-team-season records	1,709 player-team-season records	Primary defensive-front room restricted to PFF ED/DI classifications; team-season room metrics aggregate to the 128 defense-seasons.	**Complete**
Layer 2 room panel	Defensive-front team-season	2022–2025	32 defenses × 4 seasons	128 room team-seasons	Room measures constructed from qualifying ED/DI player-team-season records.	**Complete**
Layer 3 lagged resource panel	Defense-front team-season	2023–2025	96 possible team-seasons	82 team-seasons	At least 80% of room pass-rush opportunity represented by cap-matched players; prior-season player quality required by construction.	**Locked primary sample**
Temporal-validation training panel	Layer 3 team-season	2023–2024	Qualifying observations from the locked Layer 3 panel	55 team-seasons	2023–2024 observations only; preprocessing parameters estimated on training data.	**Locked**
Temporal-validation holdout	Layer 3 team-season	2025	Qualifying 2025 observations from the locked Layer 3 panel	27 team-seasons	No 2025 observations used to estimate coefficients or preprocessing parameters in the retrospective validation step.	**Locked**

Appendix Table A6. Reproducibility and Analytic-Panel Audit (continued)

Frozen data lineage, sample dimensions, temporal separation, and integrity checks for the Pressure Creation analysis

Panel C. Layer 3 reconstruction and integrity audit

Audit item	Frozen rule / expected condition	Observed result	Status	Purpose	Manifest / evidence
Fresh-environment rebuild	Reconstruct final Layer 3 variables from retained canonical inputs without inherited notebook objects.	Player prior quality, within-room resource shares, room quality, alignment, unproven share, concentration, and lagged pressure were rebuilt.	**Passed**	Tests whether final results depend on transient notebook state.	Rebuild described in final manuscript and reproducibility section.
Team-season uniqueness	Exactly one row per qualifying team-season in the final Layer 3 panel.	82 unique team-seasons.	**Passed**	Prevents duplicated organizational observations.	Final panel independently reloaded and audited.
Cap-coverage threshold	Opportunity-weighted cap coverage ≥ 80% in the primary Layer 3 sample.	82 team-seasons satisfy the locked primary rule.	**Passed**	Limits resource inference to rooms with sufficiently observed cap commitment.	Coverage threshold recorded in frozen methodology and manifest.
Bounded share variables	Within-room resource shares and related share variables remain within logical bounds.	Final panel audited for bounded shares.	**Passed**	Detects resource-allocation construction errors.	Machine-readable manifest / final panel audit.
Prior-quality missingness	Players without a valid previous-season creation estimate receive a neutral 50th-percentile prior; their current resource share is tracked separately as unproven investment.	Missing priors treated under the frozen neutral-prior rule; unproven share retained as a separate variable.	**Passed**	Separates missing prior information from an assumption of low player quality.	Transformation rule recorded in manifest.
Temporal split	Locked 82-row panel must partition into 55 training observations and 27 2025 holdout observations.	55 + 27 = 82.	**Passed**	Confirms the retrospective validation panel matches the final explanatory sample.	Sample dimensions recorded in manifest.
Validation leakage control	No 2025 observations used for validation-model coefficient estimation or preprocessing.	All preprocessing parameters estimated on the 2023–2024 training sample; 2025 reserved for evaluation.	**Passed**	Preserves the intended pseudo-out-of-sample ordering.	Final empirical-strategy specification.
FTN–PFF measurement boundary	Do not force provider pressure totals to match one-for-one.	Provider-specific measures retained in their assigned analytical roles.	**Passed**	Avoids false reconciliation when one pressured dropback can generate multiple player credits and provider definitions differ.	Measurement/reproducibility section.
Source / transformation provenance	Record input paths, checksums, transformation rules, missing-prior treatment, and sample dimensions.	Recorded in a machine-readable manifest.	**Recorded**	Allows reconstruction of the final retained panel from frozen inputs.	Manifest referenced by final manuscript.
Excluded exploratory intermediate	Final evidence must be reproducible from retained inputs.	One earlier exploratory intermediate object could not be reconstructed exactly because it was not retained; estimates dependent on it were excluded from final evidence.	**Excluded from evidence**	Prevents unreproducible exploratory results from entering the final claim set.	Explicitly disclosed in final manuscript.

	A	B	C
1	**Appendix Table A6. Reproducibility and Analytic-Panel Audit (continued)**		
2	*Frozen data lineage, sample dimensions, temporal separation, and integrity checks for the Pressure Creation analysis*		
3			
4	**Panel D. Reproducibility boundary**		
5	**Item**	**Treatment in final analysis**	**Interpretive consequence**
6	Observational design	Layers 1–2 are same-season descriptive/associational; Layer 3 uses lagged prior quality but is not a natural experiment.	Temporal ordering strengthens interpretation but does not establish a causal effect of resources or roster construction.
7	Provider definitions	FTN and PFF measures are used consistently within their assigned roles rather than treated as interchangeable physical quantities.	Results are reproducible under the frozen provider definitions but should not be interpreted as provider-invariant measurement.
8	Retrospective holdout	2025 was excluded from fitting and preprocessing during the validation exercise, but the broader variable framework was developed with knowledge of the full study period.	The exercise is retrospective / pseudo-out-of-sample validation rather than a fully prospective forecast test.
9	Canonical evidence rule	Only estimates reconstructible from retained frozen inputs are eligible for the final manuscript evidence set.	Non-reconstructible exploratory results are omitted rather than reverse-engineered or approximated.
10			
11			
12	**Notes**		
13	*1 All source and derived project data used for final estimation were frozen on July 27, 2026.*		
14	*2 Layer 3 begins in 2023 because the principal room-quality variables are lagged from the previous season.*		
15	*3 The 80% Layer 3 coverage rule is opportunity-weighted: coverage is evaluated against the pass-rush opportunities represented by cap-matched players, not simply the count of matched players.*		
16	*4 The final canonical Layer 2 player-quality table contains 1,709 player-team-season records. Earlier planning or intermediate files should not be used to override the final frozen manuscript count.*		
17	*5 The manifest is described by the final manuscript as recording source paths, checksums, transformation rules, missing-prior treatment, and sample dimensions. Individual checksum strings are not reproduced in this table.*		
18	*6 Source basis: frozen project methods, final August 2026 stitched manuscript, citation/reference audit, and retained Layer 3 methodology outputs.*		